

Variant calling & Annotation

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theory material



hands-on exercises



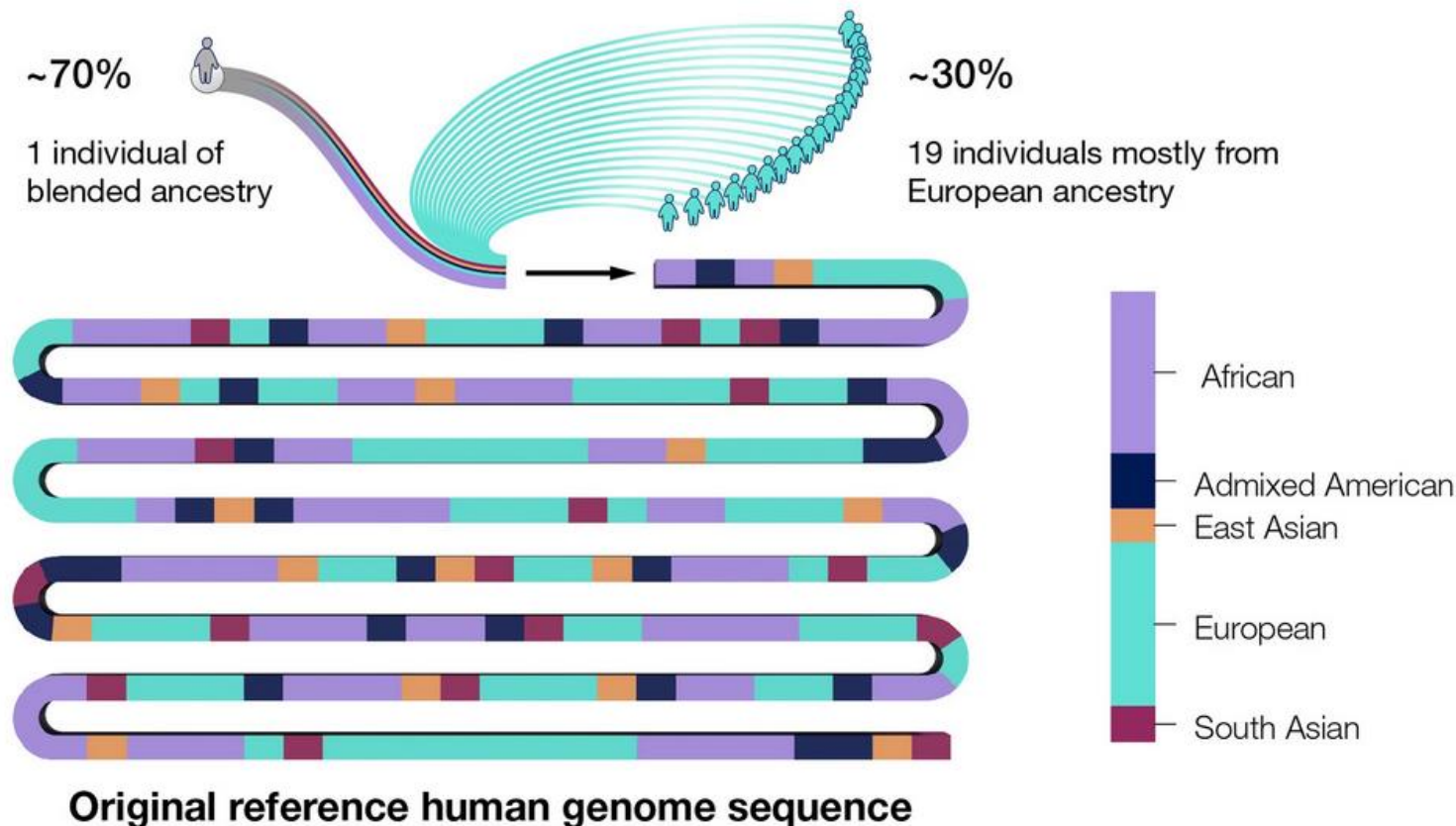
Learning Goals

- Understand the variant calling pipeline from raw sequencing reads to VCF
- Recognize differences between major variant callers and their assumptions.
- Be able to interpret a VCF file.
- Understand how to annotate variants and link them to biology/disease.
- Know limitations, pitfalls, and how to critically evaluate results.



The Reference Genome: A Standardized Map

The reference genome serves as a **universal map** for comparing genomic sequences. It is a composite, haploid sequence representing the most common alleles across a population. It's crucial to understand that it's not a single individual's genome but a standardized template that provides a common coordinate system.



Detecting variants in a person's genome sequence usually requires an existing sequence for comparison — a *reference*.

A reference [human genome](#) sequence is an established, high-quality and well-accepted sequence of a human genome (i.e., one sequence for each of the 23 human [chromosomes](#)).

A reference human genome sequence is not an actual sequence from one individual but pieced together from multiple people. The important feature of a reference human genome sequence is that it depicts one assigned nucleotide for every position across the human genome. In this regard, a reference human genome sequence only represents the sequence of one copy of each [chromosome](#), whereas a person's genome sequence contains the sequences of both copies of each chromosome (i.e., people are [diploid](#)).

The Reference Genome: A Standardized Map

- **The Human Genome Reference Sequence:**
 - **Current Standard (de-facto): hg19 / hg37** : These two versions are functionally the same and are still very widely used, especially in older datasets and clinical pipelines. Many established tools and databases (like dbSNP and 1000 Genomes) are built on this assembly.
 - **Newer Standard: hg38**: This is the current, more complete reference assembly. It includes improvements such as more accurate centromeric regions and the incorporation of "alternative loci" to better represent human genetic diversity. All major consortia (like the Genome Aggregation Database, gnomAD) now use hg38. The transition from hg19 to hg38 requires a process called liftover, which can be a significant technical challenge.
- **Other Organisms:** Not all organisms have a fully assembled reference genome.



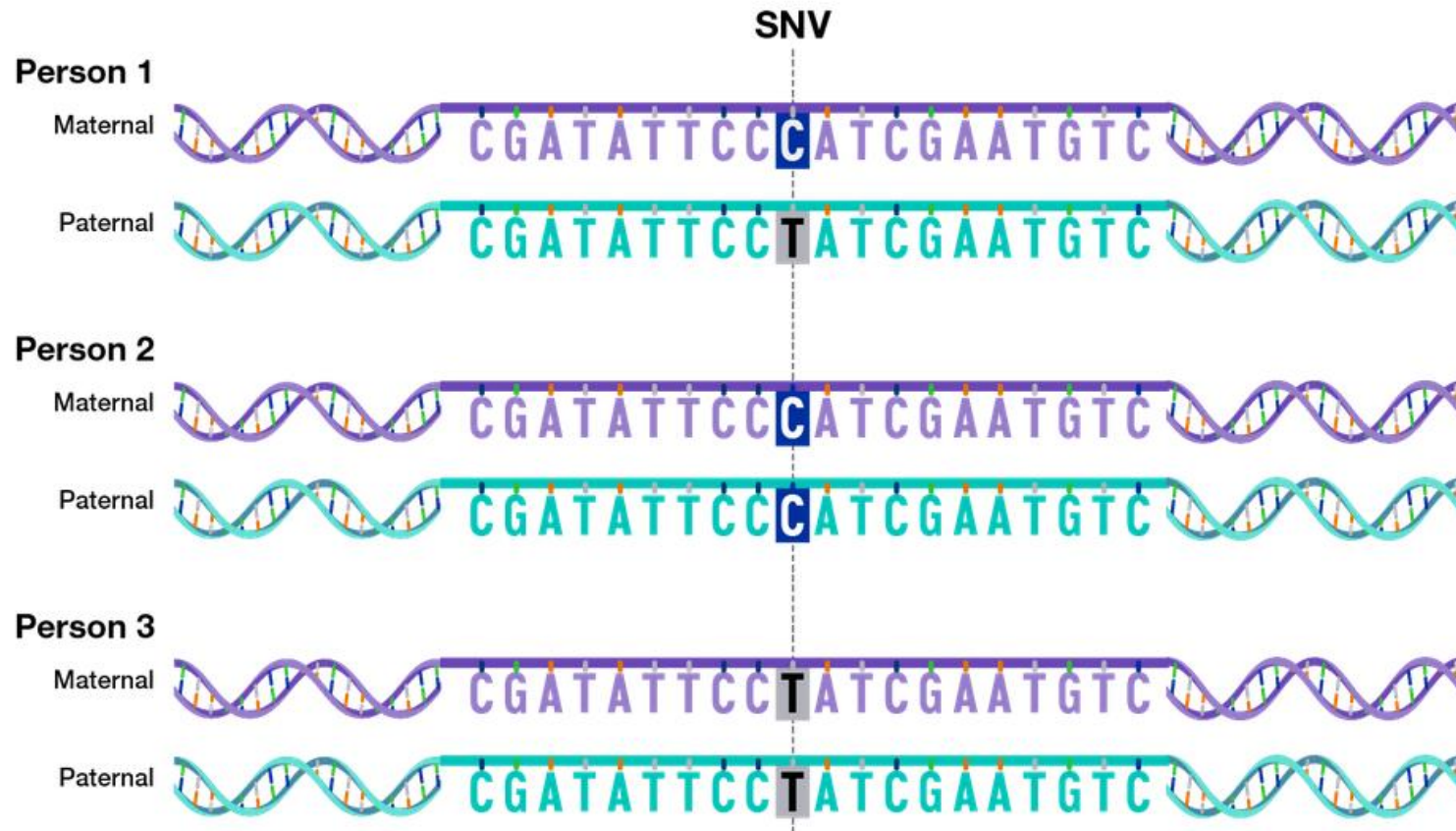
What Is Human Genomic Variation?

The vast majority of the DNA letters in peoples' genomes is identical, but a small fraction of those letters varies. This [genomic variation](#) accounts for some of the differences among people, including important aspects of their health and susceptibility to diseases.

- [Genomic variation](#) reflects the differences in a person's DNA compared to other peoples' DNA.
- There are multiple types of variants in human genomes, ranging from small differences to large differences.
- A very small subset of genomic variants contributes to human health and disease.
- Researchers create reference [human genome](#) sequences to help detect genomic variants in each sequenced human genome.
- On average, a person's genome sequence is ~99.6% identical to a reference [human genome](#) sequence; that person's set of genomic variants accounts for the ~0.4% difference.
- The human pangenome is a more comprehensive framework that aims to account for [genomic variation](#) across human populations, thereby reducing biases that can come with the use of a single reference [human genome](#) sequence.

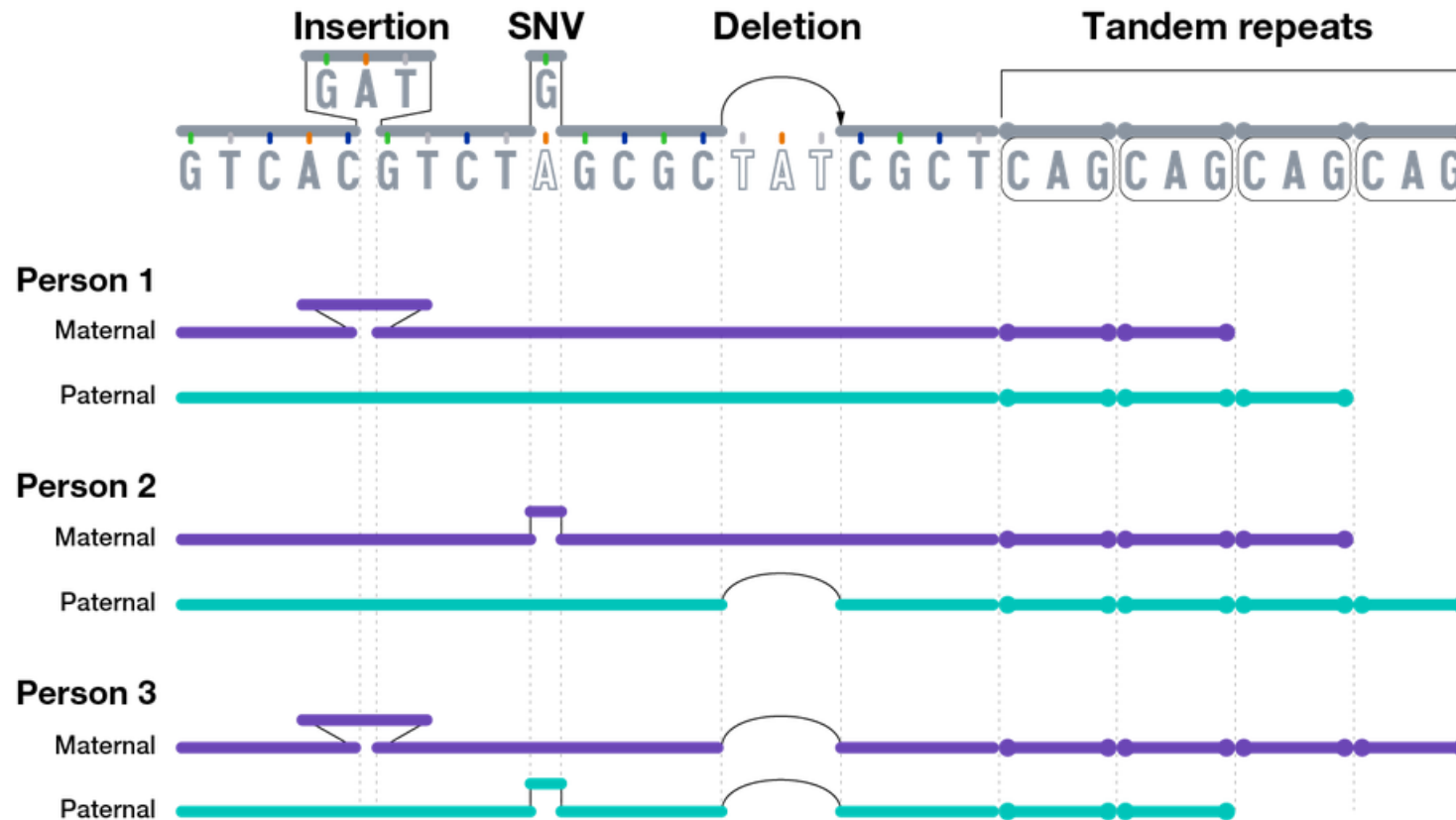


What Are the Different Types of Genomic Variants?



The smallest genomic variants are single-nucleotide variants (SNVs). Each SNV reflects a difference in a single nucleotide. For a given SNV, the DNA letter at that genomic position might be a C in one person but a T in another person. SNVs are the most common type of [genomic variation](#). A subtype of SNVs is called a single-nucleotide polymorphism (SNP; pronounced "snip"). To be considered a SNP, a SNV must be present in **at least 1% of the human population**. As such, SNV is a more general term that includes both relatively common (such as SNPs) and rare single-nucleotide differences. For simplicity, we refer to all single-nucleotide differences as SNVs, regardless of their relative frequency.

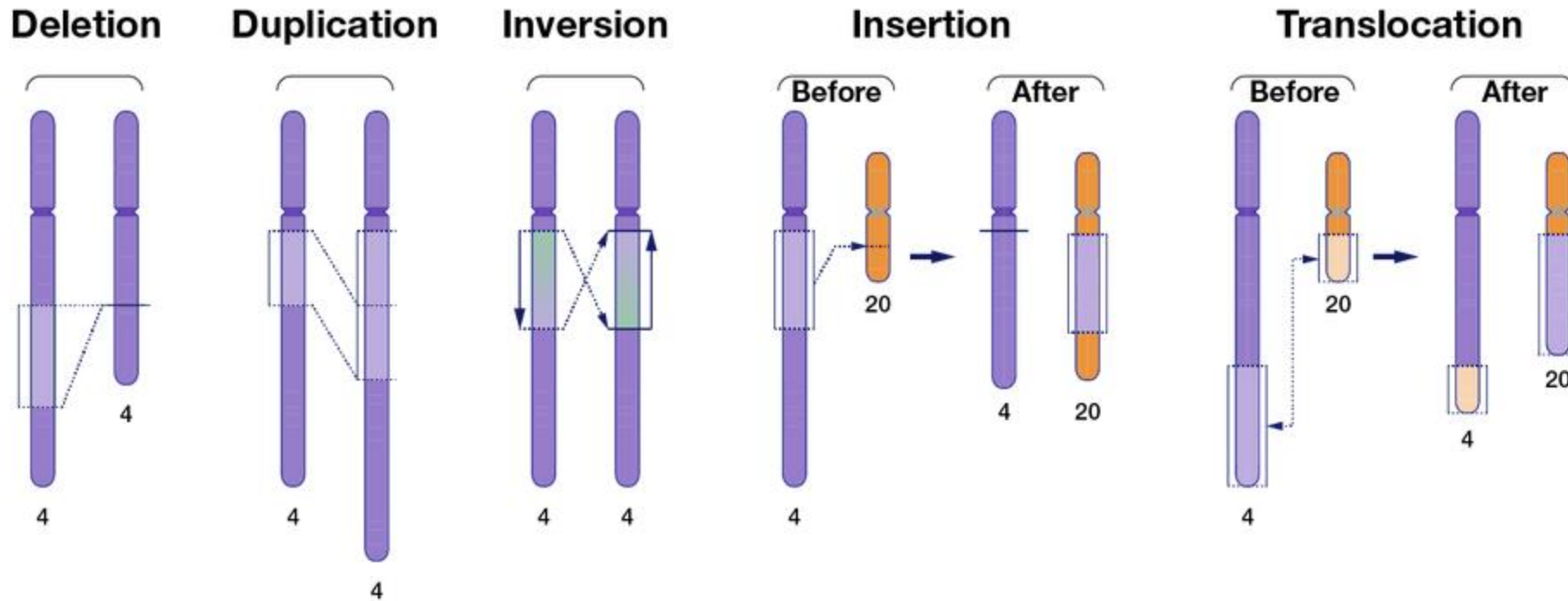
What Are the Different Types of Genomic Variants?



Another group of small genomic variants are insertions and deletions (*indels*). Insertion/deletion variants reflect extra or missing DNA nucleotides in the genome, respectively, and typically involve fewer than 50 nucleotides. Insertion/deletion variants are less frequent than SNVs but can sometimes have a larger impact on health and disease.

One of the most common types of insertion/deletion variants are tandem repeats (also known as [microsatellites](#)). Tandem repeats are short stretches of nucleotides that are repeated multiple times and are highly variable among people. Different [chromosomes](#) can vary in the number of times such short nucleotide stretches are repeated, ranging from a few times to hundreds of times.

What Are the Different Types of Genomic Variants?



[Genomic variation](#) also extends beyond small stretches of nucleotides to larger chromosomal regions. These large-scale genomic differences are called structural variants and involve at least 50 nucleotides and as many as thousands of nucleotides that have been inserted, deleted, inverted or moved from one part of the genome to another. Tandem repeats that contain more than 50 nucleotides are considered structural variants; in fact, such large tandem repeats account for nearly half of the structural variants present in human genomes. When a structural variant reflects differences in the total number of nucleotides involved, it is called a copy-number variant (CNV). Note that CNVs are distinguished from other structural variants, such as inversions and translocations, because the latter types often do not involve a difference in the total number of nucleotides.

Comparing individual human genome sequences

**Reference
genome sequence**

[illegible]

TTAACTTGAGTGGATTACTAACGGTATTTCTGGGATTAAA.....GTATAGTGTGAGTATCGGAGA

TTAAATTGAGTGGTTTAATAACGGTATGTCTGGGATTAAAGTGTAGTATAGTGTGATTATCGGAGA



**Your
genome sequence**

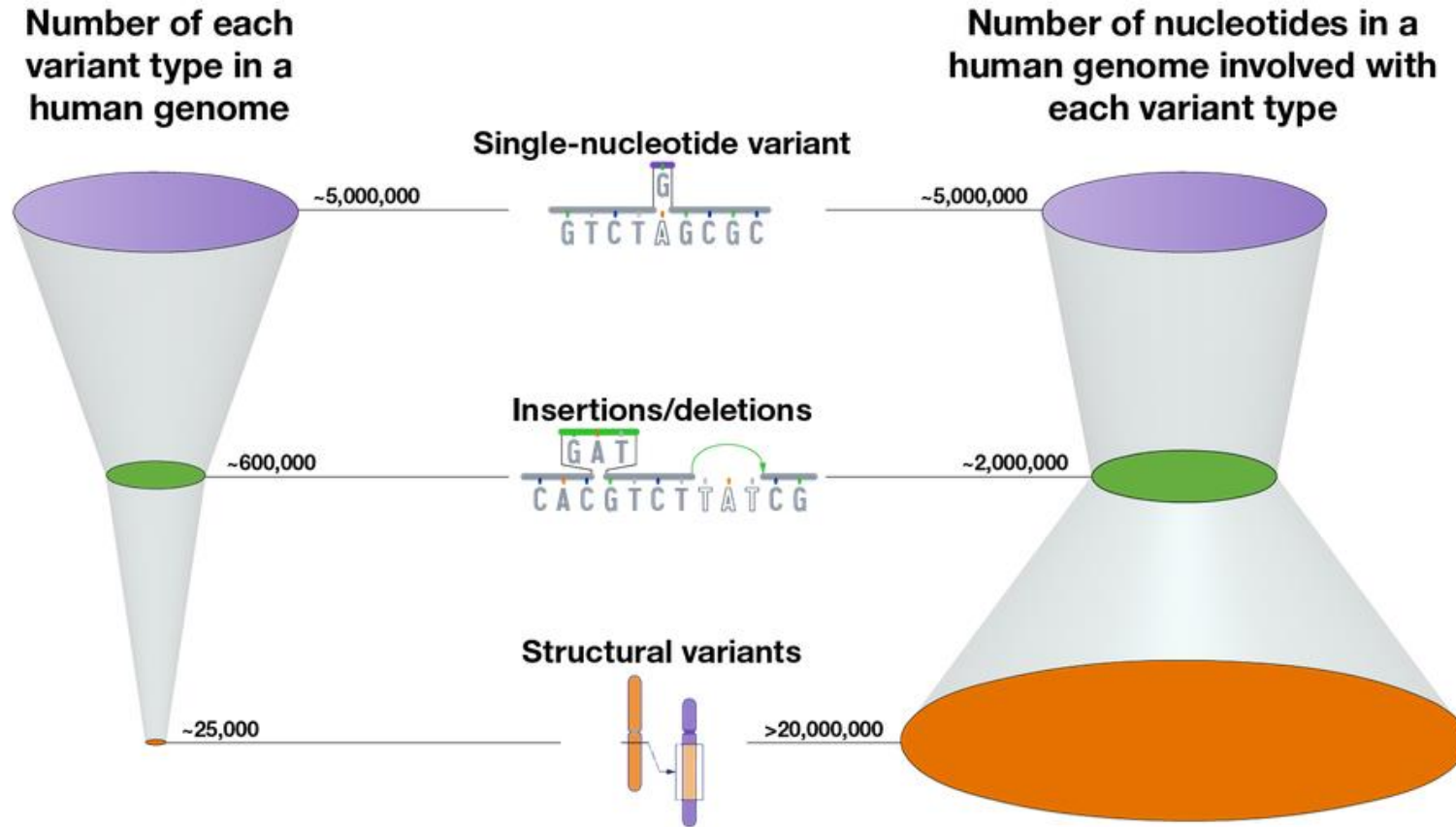
[illegible]

TAGTGTGAGTATCGGAGA

TAGTGTGATTATCGGAGA

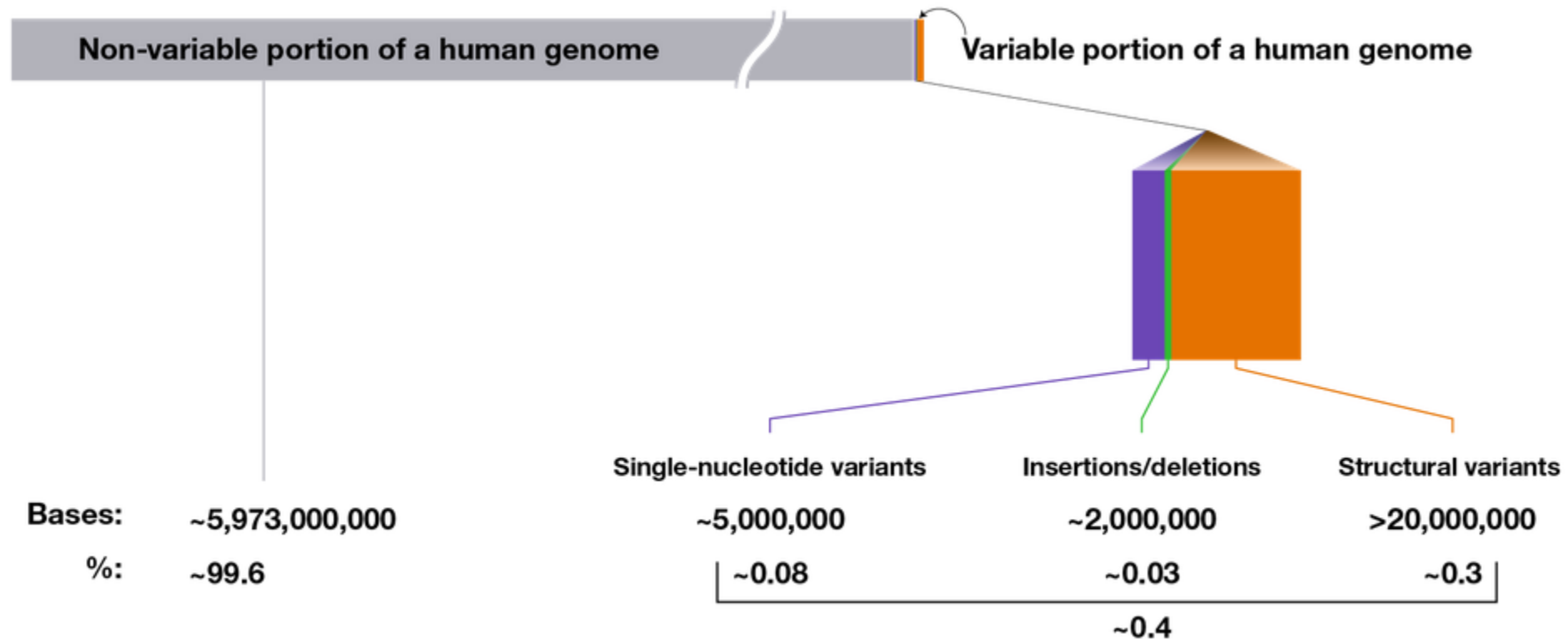
TTAACTTGAGTGGATTACTAACGGTATTTCTGGGATTA...GTATAGTGTGAGTATCGGAGA

What Is the Inventory of Genomic Variants in a Typical Human Genome Sequence?



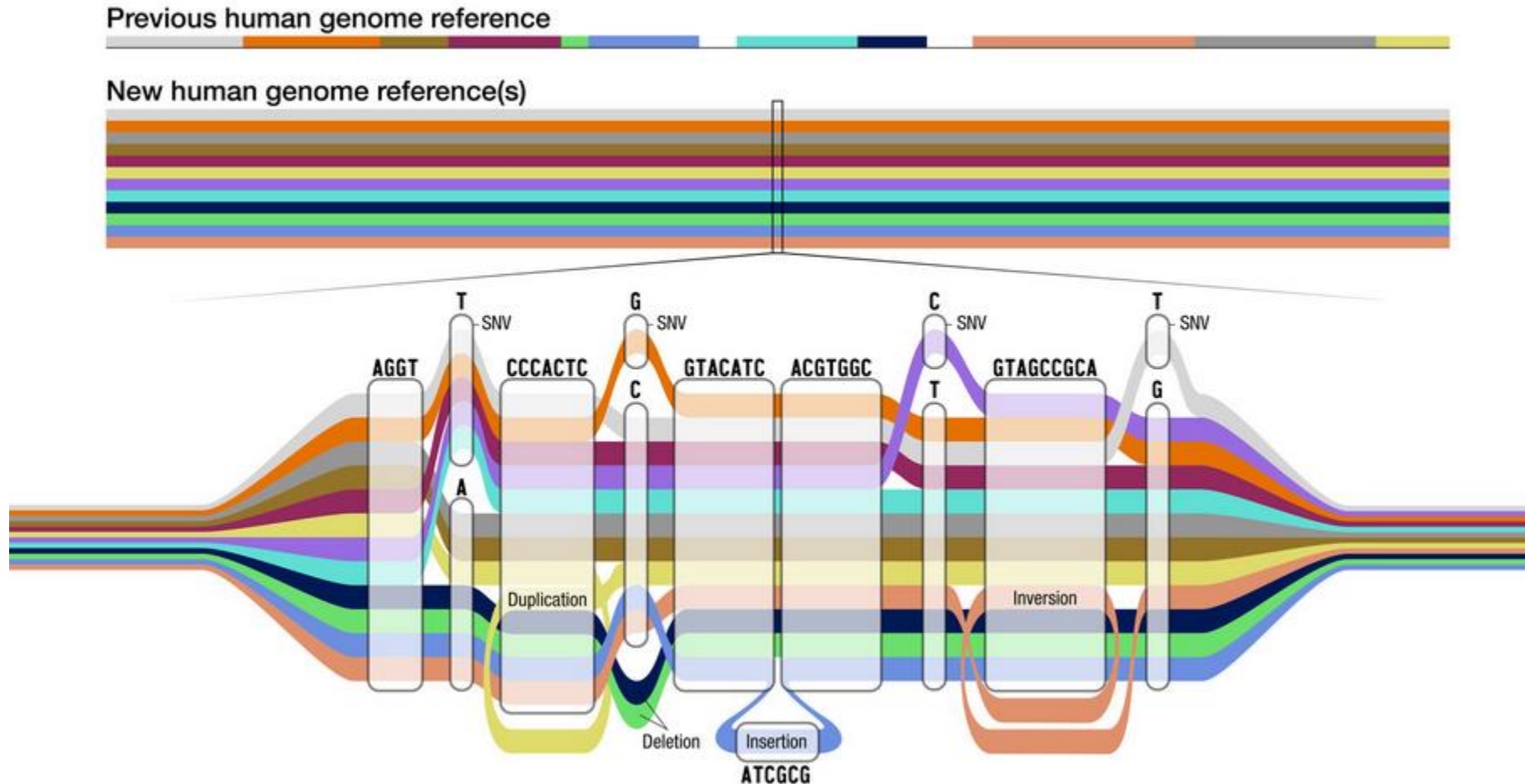
- What does a typical [human genome](#) sequence look like with respect to the variants that it contains? On average, compared to a reference human genome, a person's ~6 billion-nucleotide genome sequence will have:
- ~5,000,000 SNVs that involve ~5,000,000 nucleotides
- ~600,000 insertion/deletion variants that involve ~2,000,000 nucleotides
- ~25,000 structural variants that involve >20,000,000 nucleotides.
- These numbers represent current estimated averages and will likely change as more human genomes are sequenced to completion. Furthermore, multiple approaches can be used to calculate how the different types of genomic variants account for the differences among genomes. In fact, calculating the total number of nucleotides involved in structural variants in each [human genome](#) is quite complicated and remains an active area of research.

What Is the Inventory of Genomic Variants in a Typical Human Genome Sequence?



On average, a [human genome](#) sequence (consisting of ~6,000,000,000 nucleotides) is ~99.6% identical — or non-variable — compared to a reference human genome sequence. The variable portion of a human genome sequence totals ~0.4%, which consists of the indicated contributions from each of the three major types of genomic variants. These same general numbers are also found when directly comparing two peoples' genome sequences (as opposed to comparing to a reference human genome sequence).

What Is a Pangenome Reference Sequence and How Is It Used?



Reference [human genome](#) sequences are invaluable for detecting genomic variants in each newly sequenced human genome. But the currently available reference human genome sequences do not accurately represent the genomic diversity of the human population, and that lack of diversity can introduce biases when analyzing some peoples' genome sequences.

To address this deficiency, researchers are working to generate a more complete set of reference [human genome](#) sequences that better reflect all of humanity. A "pangenome" is the collective genome sequences of multiple individuals that better represents the genomic diversity of the species. The human pangenome reference sequence will provide a better tool for comparing genome sequences from people all over the world in an effort to detect and characterize genomic variants more completely, including those with important roles in health and disease.

Why do we need a new human pangenome reference?



The original human genome reference sequence is outdated.

The original human genome reference sequence was generated by the Human Genome Project in 2003. While this reference sequence has been regularly updated as researchers fixed errors and filled in missing regions of the genome, it only reflected data generated from about 20 people. Most of that first human genome reference sequence was just from one person.



A human pangenome reference better reflects human diversity.

The new human pangenome reference includes genomic data from 47 people who are collectively more globally diverse. Researchers expect that number to reach 350 people by 2024. A human pangenome reference that better reflects genomic variation across all human populations will help ensure that it is beneficial for everyone and that genomics will advance in an equitable way.

The previous human genome reference sequence has missing pieces (i.e., gaps).

The previous human genome reference sequence is only 92% complete, with an estimated 8% of the human genome missing because of gaps in the sequence. Recently, new types of DNA sequencing technologies have helped researchers read longer stretches of DNA at a time, allowing them to fill in missing sequences within those gaps, especially in areas that were repetitive and harder to read. The new human pangenome reference is more comprehensive and incorporates the missing 8% of the human genome sequence, adding over 100 million new bases.

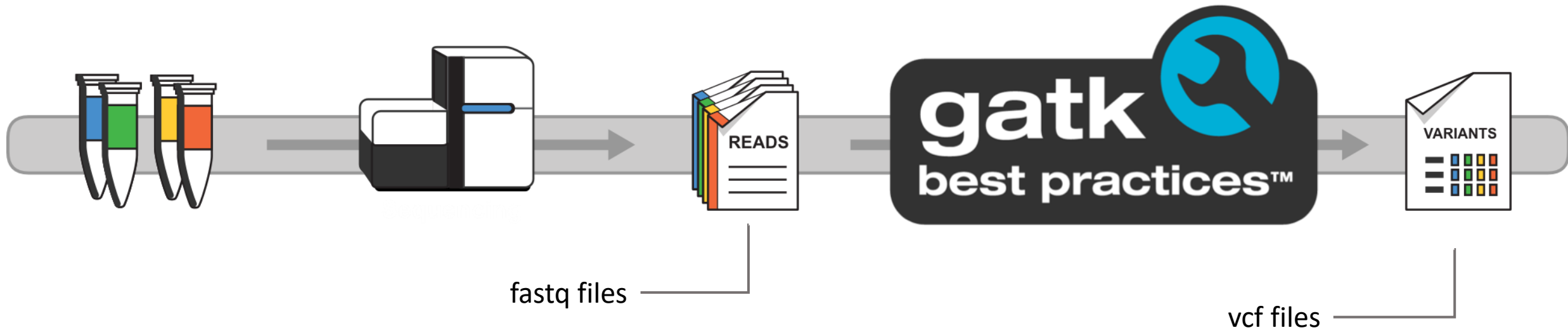


The new human pangenome reference can help researchers make more genomic discoveries.

The new human pangenome reference will help researchers analyze human genome sequences and understand how genomic variants influence human health. Using the human pangenome reference, researchers are already finding genomic variants that were not previously identified using the older human genome reference sequences.



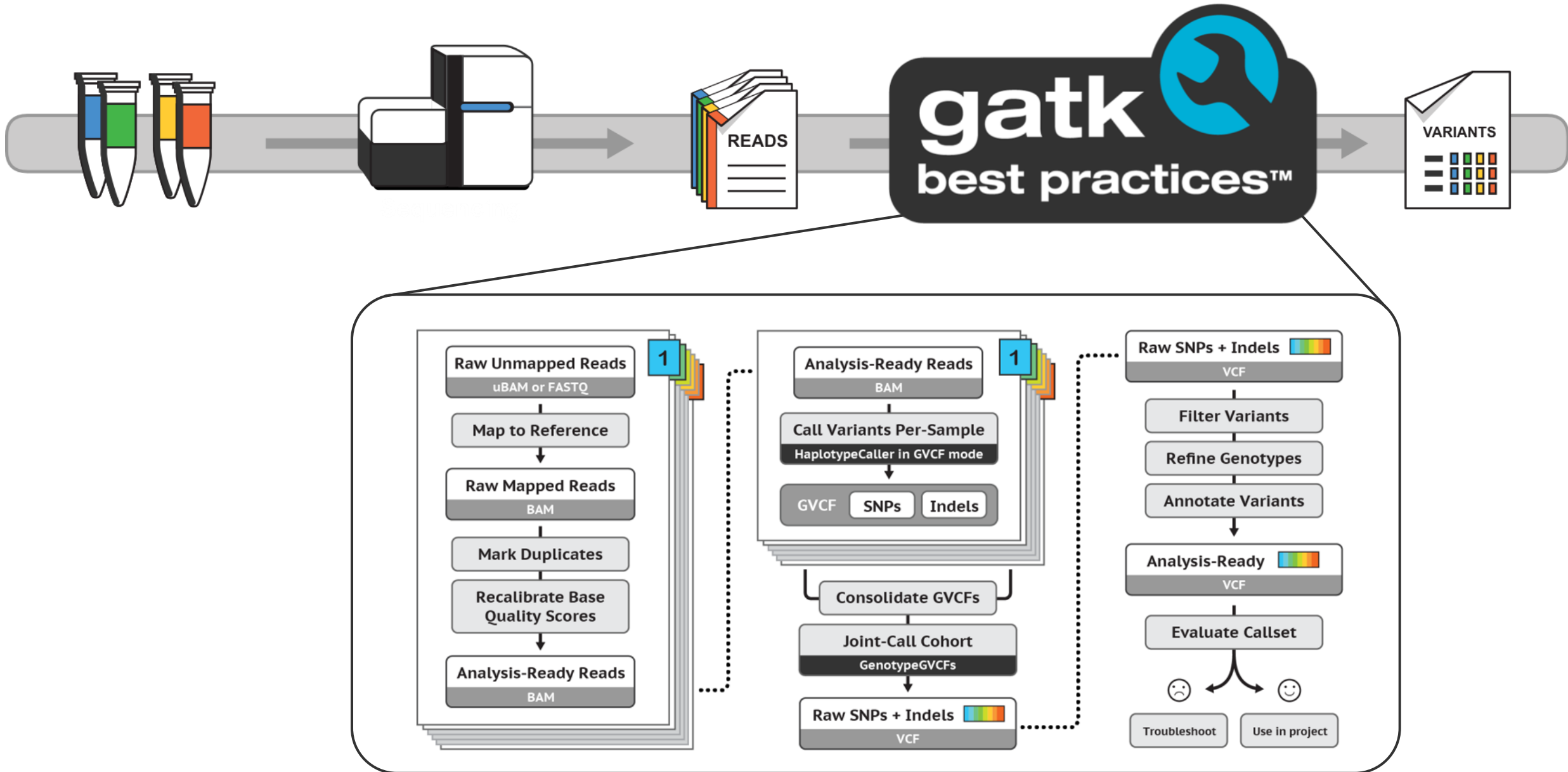
Variant discovery : identify variants in sequencing data



Variant calling pipeline from raw sequencing reads to VCF

Variant calling is the process of detecting differences between a sample genome and a reference genome

Variant discovery : identify variants in sequencing data



Variant discovery : identify variants in sequencing data

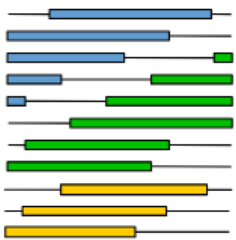


Data processing

Variant discovery

Callset refinement

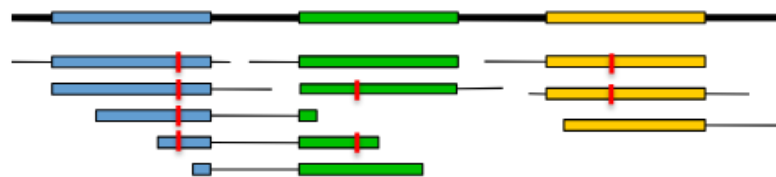
FASTQ



Enormous pile of short reads

Sequence + base qualities

SAM/BAM



Alignment map

VCF

- ■ site 1 description + sample genotypes
- ■ site 2 description + sample genotypes
- ■ site 3 description + sample genotypes

Site-by-site genomic variation

Reference genome

Mapped reads

Understanding the Landscape of Genetic Variation

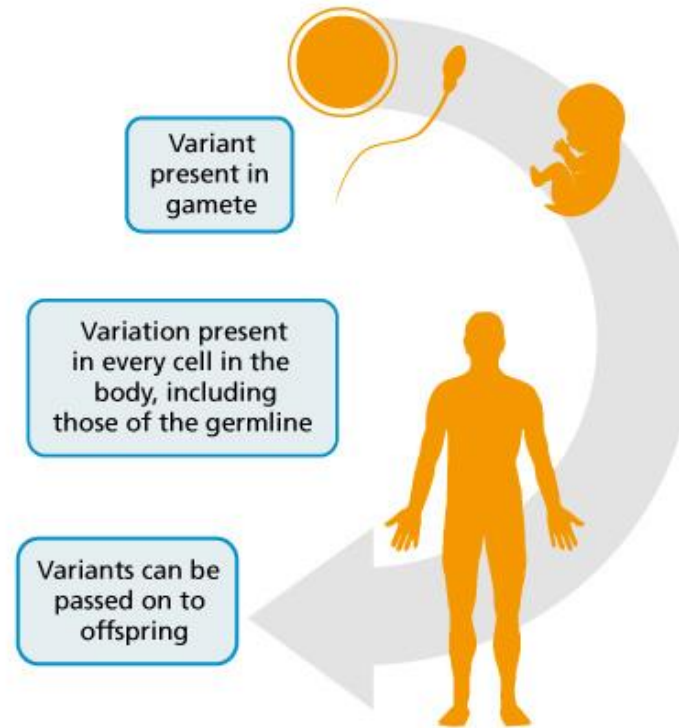
Germline vs. Somatic Variants

- **Germline Variants**

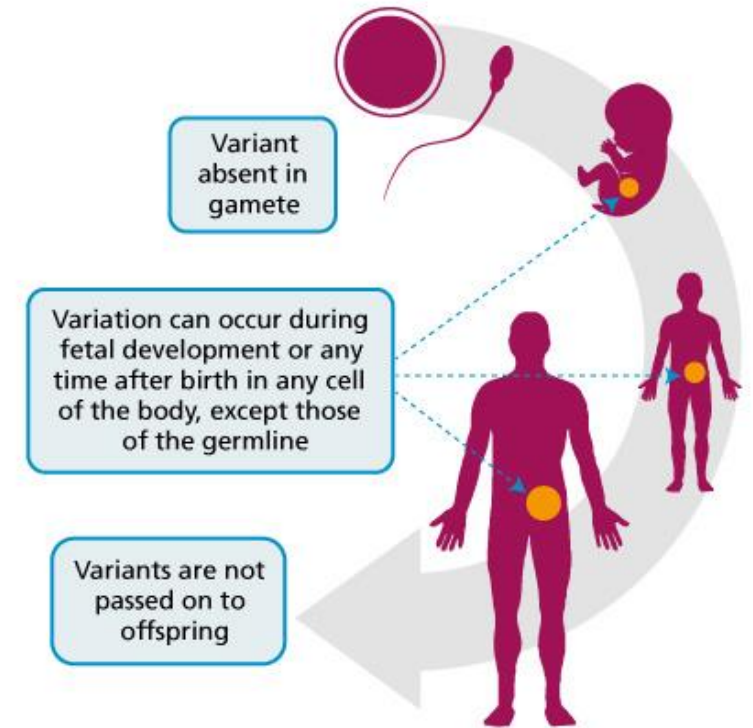
These are inherited from a parent and are present in every cell of an individual's body. They arise from genetic mutations in the egg or sperm cells. Calling germline variants is a key component of understanding inherited disease risk, population genetics, and human evolution. For example, a single nucleotide polymorphism (SNP) associated with a higher risk for a certain condition would be a germline variant.

- **Somatic Variants**

These are non-inherited genetic changes that occur after conception. They are not found in every cell of the body but are present in specific cell populations, such as a tumor. A classic example is a cancer-driving mutation like a change in the EGFR gene, which is found only in the tumor cells and not in the surrounding healthy tissue. Identifying somatic variants is the cornerstone of cancer genomics and targeted therapy.



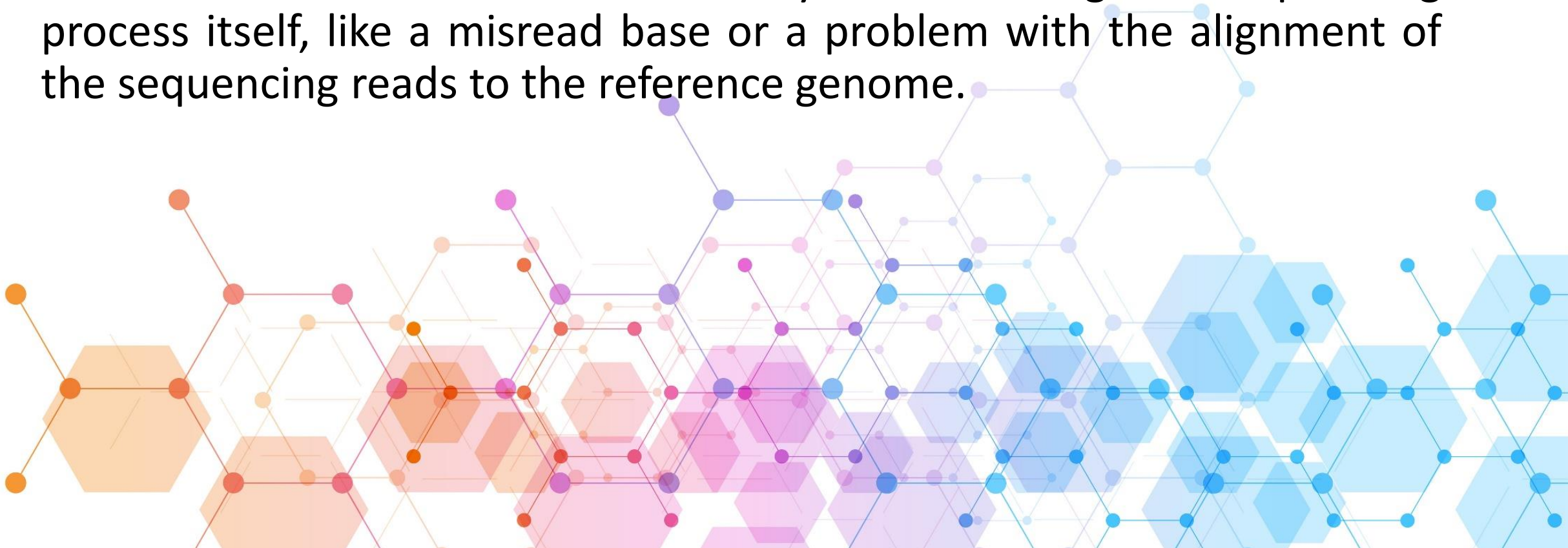
Germline variants



Somatic variants

The Challenge of Distinguishing True Variants from Artifacts

A **genetic variant** is a genuine change in the DNA sequence that is actually present in the genome of the individual or a cell population. In contrast, an **artifact** is an "observed" change in the sequence data that isn't real. These are often caused by errors during the sequencing process itself, like a misread base or a problem with the alignment of the sequencing reads to the reference genome.



Common Next-Generation Sequencing (NGS) Artifacts

- **Sequencing-Specific Artifacts**

These are errors introduced during the DNA sequencing process itself. They are often platform-specific and can appear as "variant calls" at positions that aren't actually different from the reference.

- *Low-Quality Bases: A base call with a low Phred score (e.g., $Q < 30$) has a higher chance of being a misread. These can look like a variant, but are simply an error.*
- *Read-Through or Contamination: If DNA libraries were not properly cleaned, adapter sequences can be read and incorrectly aligned, leading to false variants.*
- *Sequencing Hotspots: Certain positions or motifs (e.g., homopolymers of A or T repeats) are prone to systematic sequencing errors, causing recurrent false positives.*

- **Alignment-Specific Artifacts**

These occur when the sequencing reads are aligned to the reference genome, even if the reads themselves are high-quality.

- *Misalignment near Indels: Reads containing a small insertion or deletion (indel) can be difficult to align correctly. They might be shifted slightly, causing nearby bases to look like false single nucleotide variants.*
- *Reads Misaligned to a Parologue or Pseudogene: A read from a gene might incorrectly align to a similar-looking pseudogene (a non-functional copy of the gene) somewhere else in the genome. This makes the entire read look "different" and can cause a flood of false variant calls.*

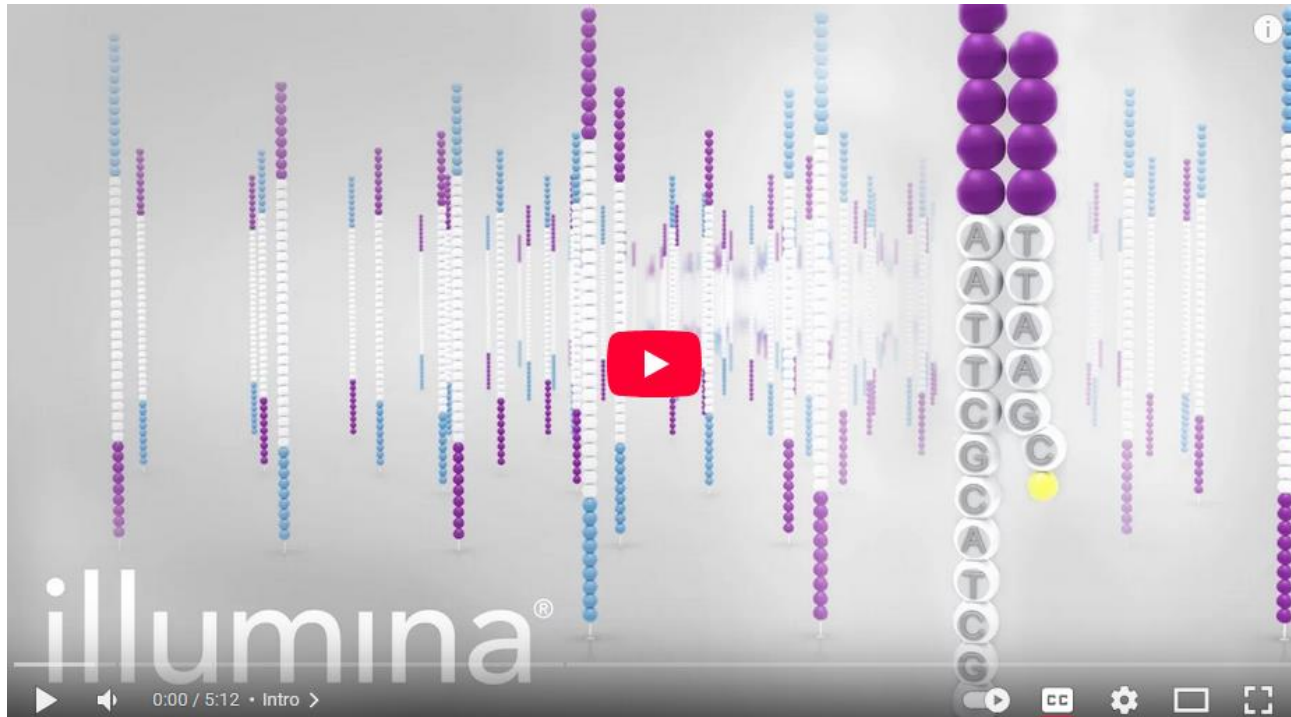
- **Structural and Copy Number Variation**

This is a more subtle source of artifacts. For example, a whole gene deletion on one chromosome might look like a 50% allele frequency change in your data. It might be mistaken for a heterozygous variant instead of a genuine copy number loss.

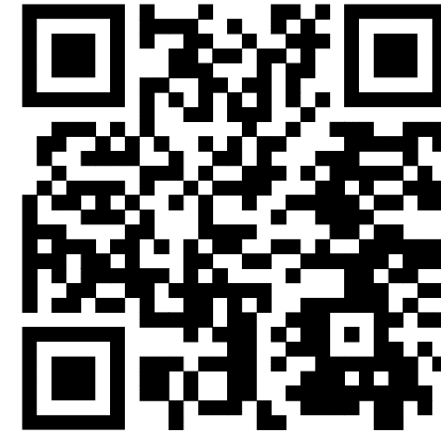
Phases of variant calling



Sequencing by synthesis

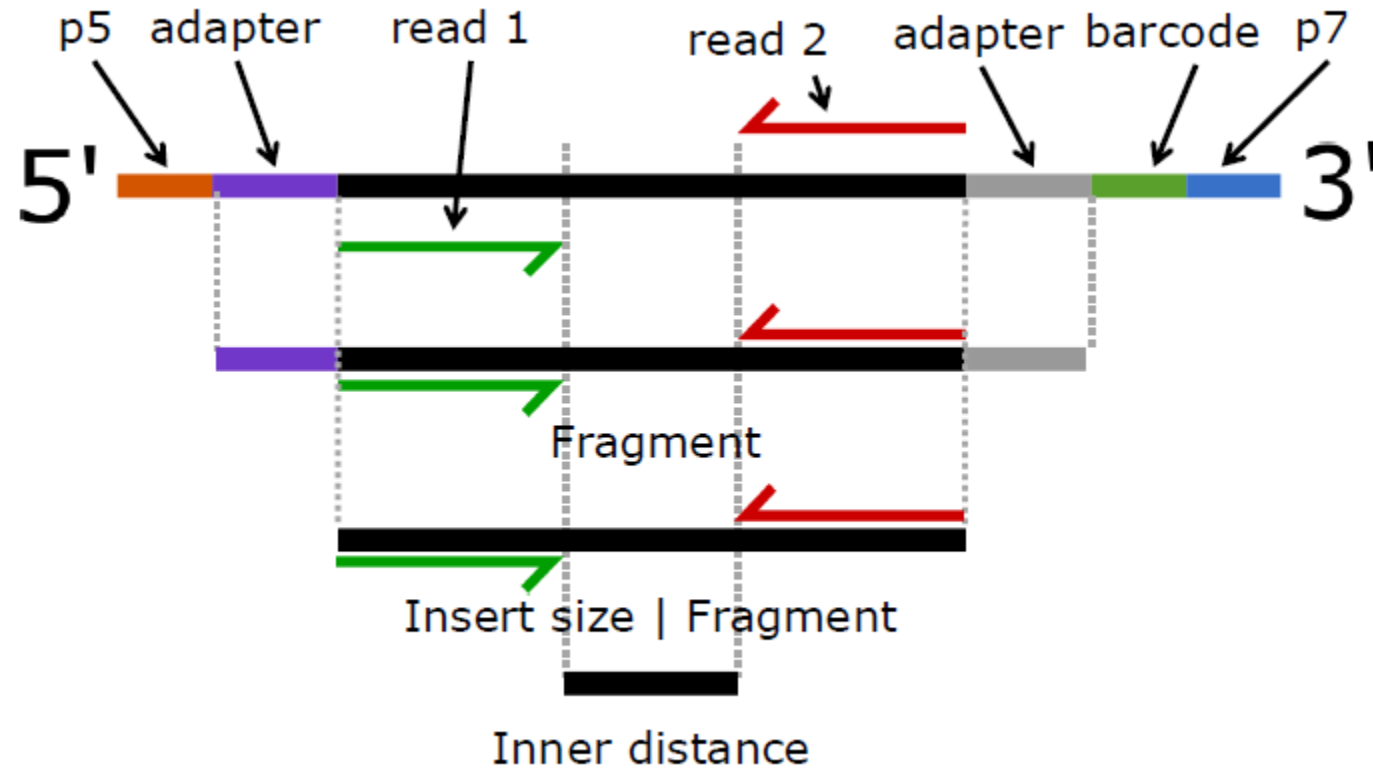


<https://www.youtube.com/watch?v=fCd6B5HRaZ8>



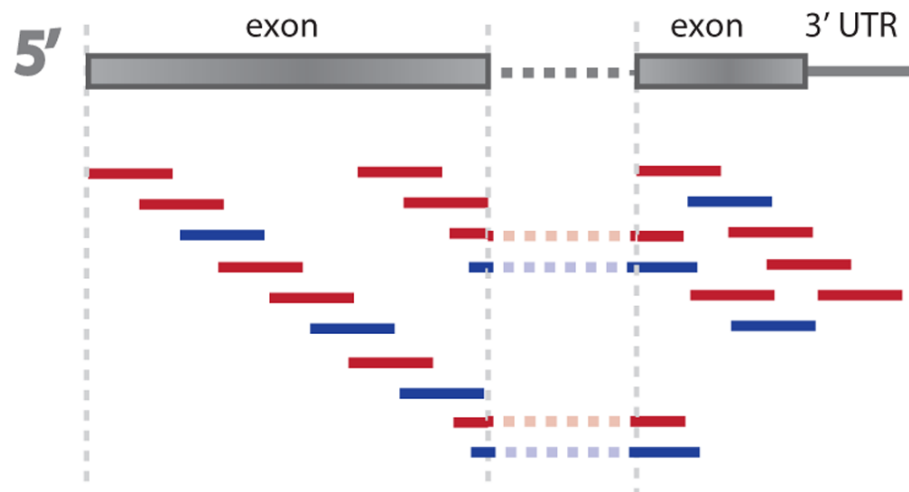
Overview of Illumina
Sequencing by Syn-
thesis

Insert Size And Fragment Size ?

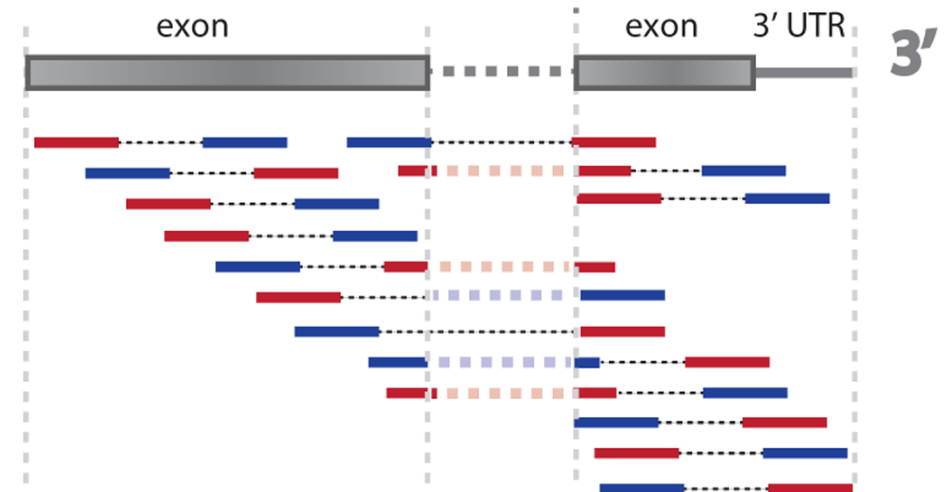


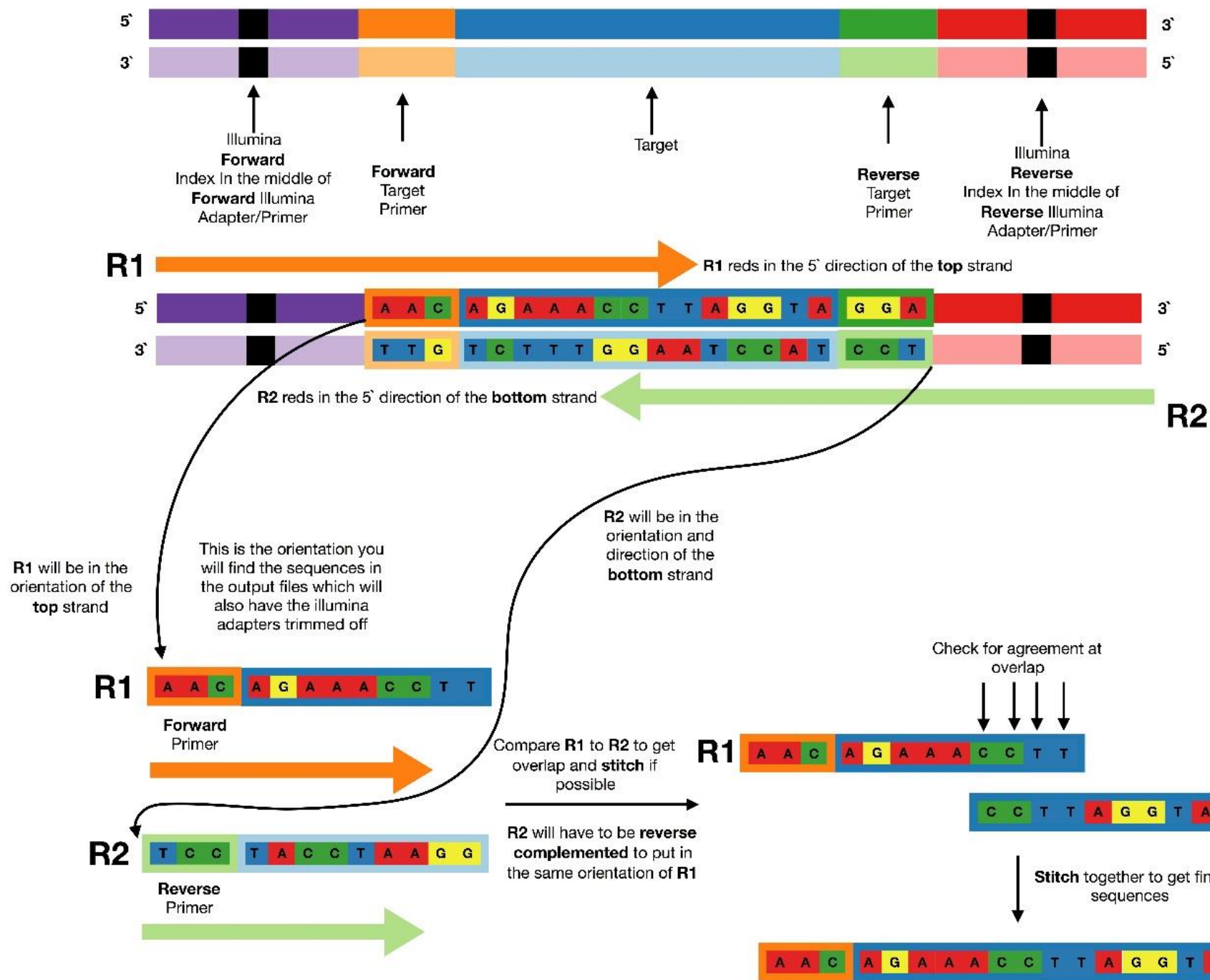
Single-end/Paired-end sequencing

Single-end sequencing

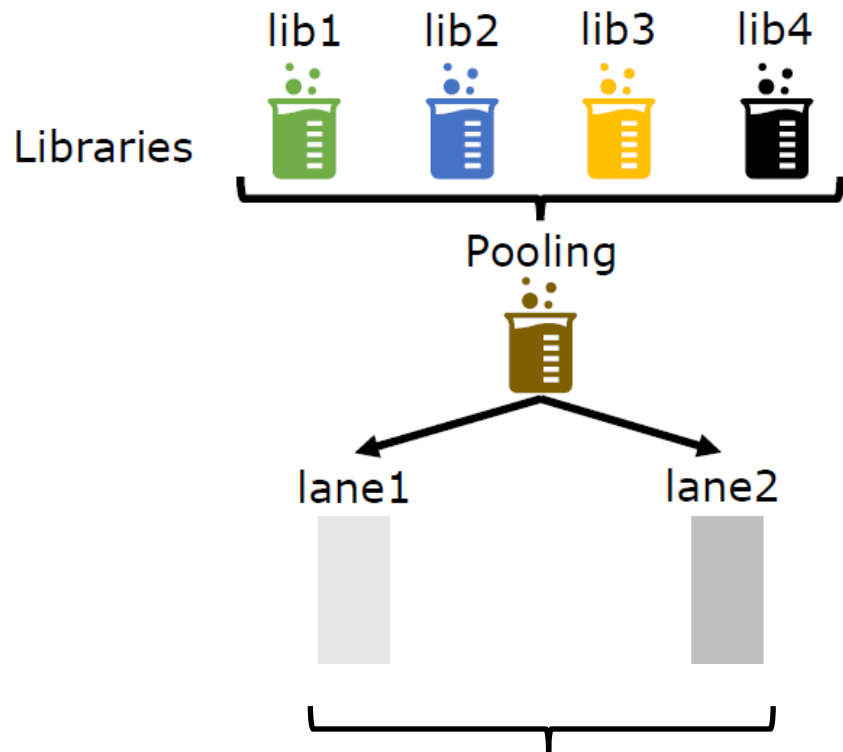


Paired-end sequencing





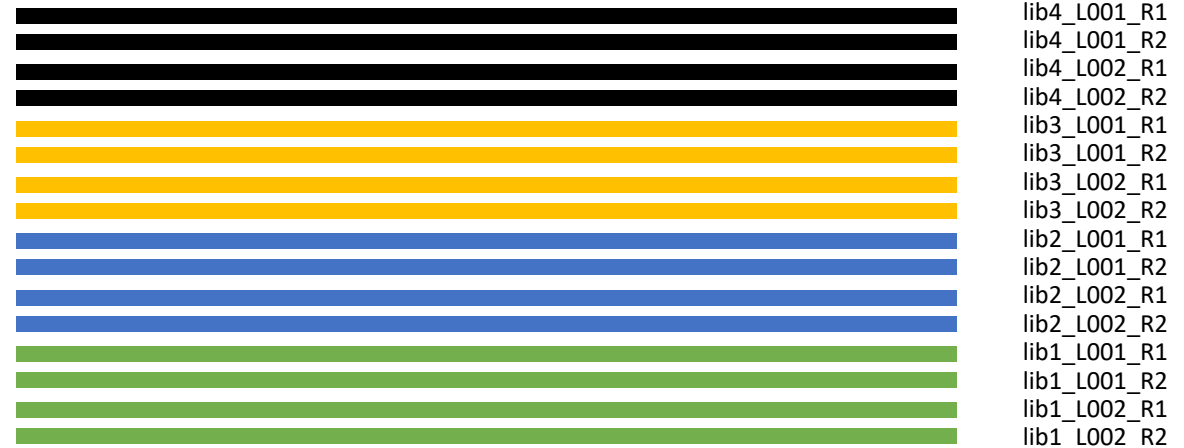
The key is that R1 and R2 come from the same physical fragment of DNA and are therefore "paired". R1 is the forward read. R2 is the reverse-complement of the other end of the fragment. When you align these reads to a reference genome, the alignment tool will automatically account for this orientation. R1 will align in the forward orientation and R2 will align in the reverse orientation, with the two reads pointing toward each other. The distance between them on the reference genome is known as the insert size, which is a crucial quality metric for the sequencing run.



Demultiplexing

Library: fragments from one (c)DNA sample that share a barcode

- **Sequencing run:** complete cycle of generating reads on a machine
- **Flow cell:** physical platform where sequencing reactions take place. Used once in a sequencing run.
- **Lane:** compartment within the flow cell. An Illumina flow cell often has multiple lanes (2 or 4)



1. Raw Sequencing Reads (FASTQ)

- Input from Illumina, Nanopore, or PacBio machines.
- FASTQ contains:
 - **Sequence** (ATCG...)
 - **Quality scores** (Phred scores per base).

```
Identifier | @HWI-EAS209_0006_FC706VJ:5:58:5894:21141#ATCACG/1
Sequence  | TTAATTGGTAAATAAATCTCCTAATAGCTTAGATNTTACCTTNNNNNNNNNTAGTTTCTTGAGA
+ sign & identifier | +
Quality scores | efcfffffcfeefffcfffffddf`feed]`_]_Ba_^__[YBBBBBBBBBBRTT\]][] dddd`
```

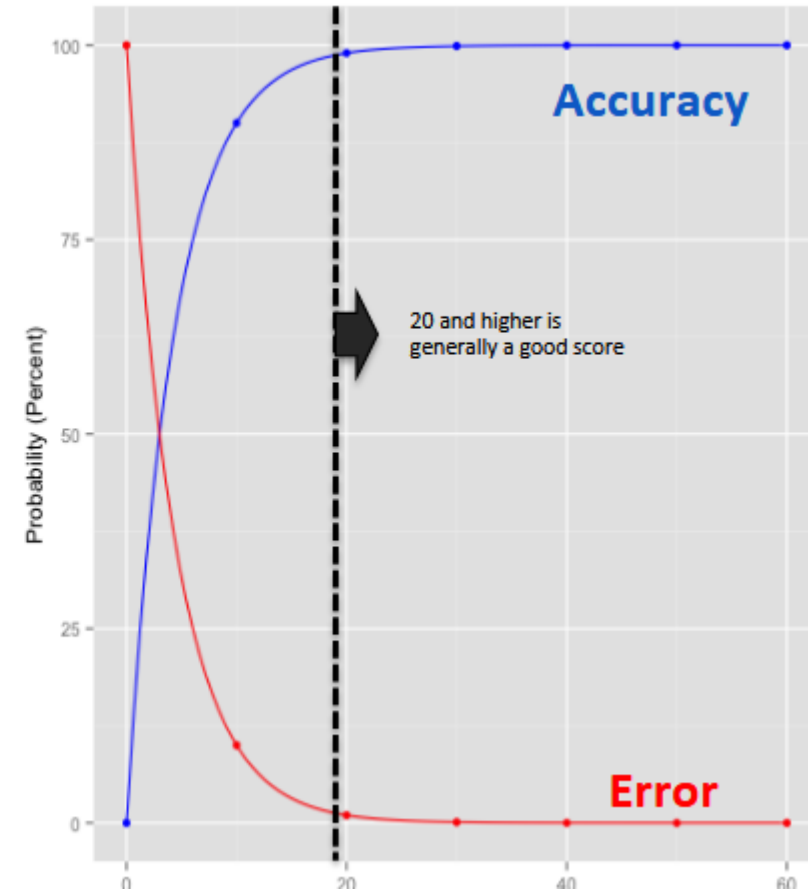
Base T
phred Quality] = 29

Quality (Phred) scores

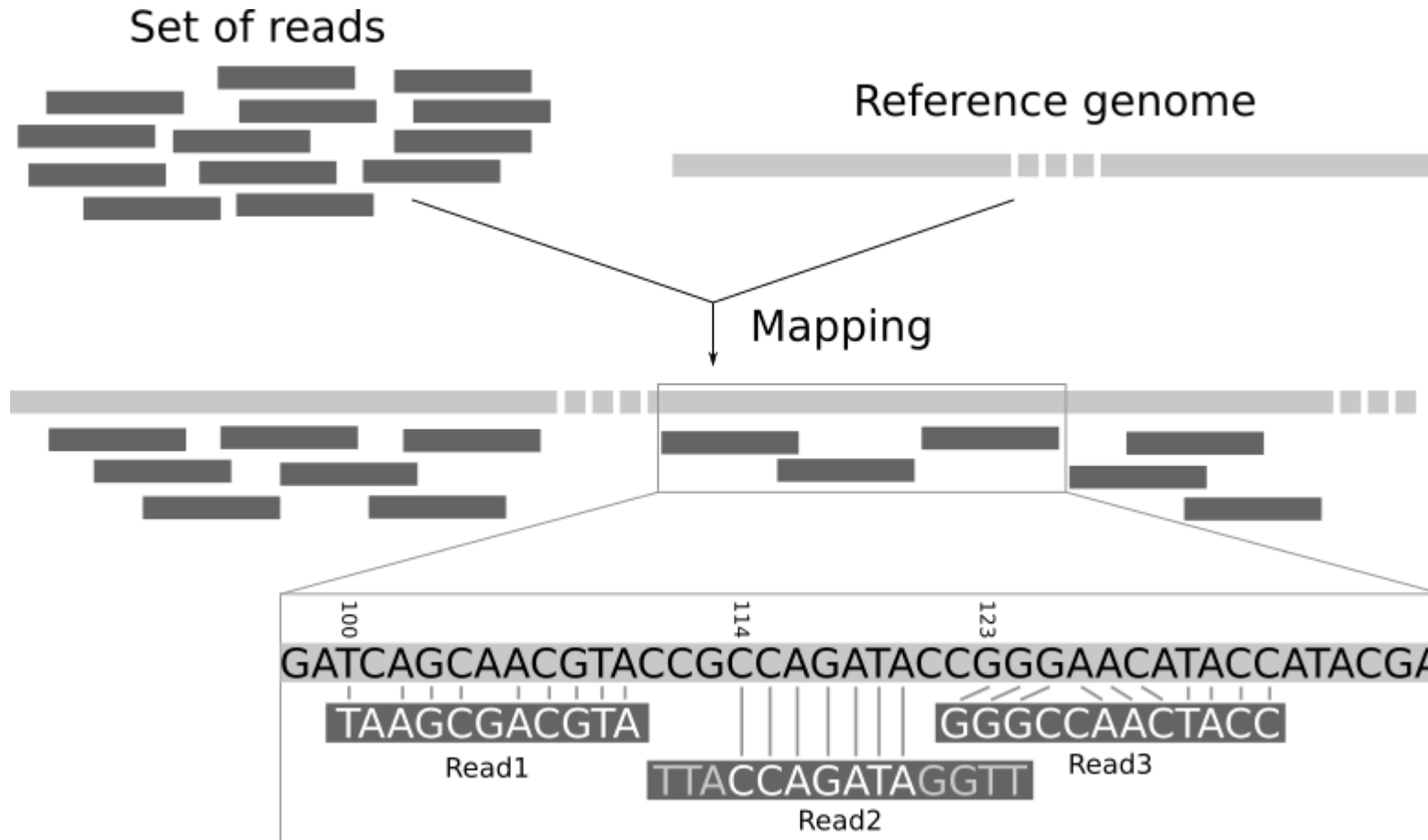
- $\text{Phred value} = -10 * \log_{10}(\epsilon)$
- Examples:
 - 90% confidence (10% error rate) = Q10
 - 99% confidence (1% error rate) = Q20
 - 99.9% confidence (.1% error rate) = Q30

ASCII_BASE=33 Illumina, Ion Torrent, PacBio and Sanger

Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII
0	1.00000	33 !	11	0.07943	44 ,	22	0.00631	55 7	33	0.00050	66 B
1	0.79433	34 "	12	0.06310	45 -	23	0.00501	56 8	34	0.00040	67 C
2	0.63096	35 #	13	0.05012	46 .	24	0.00398	57 9	35	0.00032	68 D
3	0.50119	36 \$	14	0.03981	47 /	25	0.00316	58 :	36	0.00025	69 E
4	0.39811	37 %	15	0.03162	48 0	26	0.00251	59 ;	37	0.00020	70 F
5	0.31623	38 &	16	0.02512	49 1	27	0.00200	60 <	38	0.00016	71 G
6	0.25119	39 '	17	0.01995	50 2	28	0.00158	61 =	39	0.00013	72 H
7	0.19953	40 (	18	0.01585	51 3	29	0.00126	62 >	40	0.00010	73 I
8	0.15849	41)	19	0.01259	52 4	30	0.00100	63 ?	41	0.00008	74 J
9	0.12589	42 *	20	0.01000	53 5	31	0.00079	64 @	42	0.00006	75 K
10	0.10000	43 +	21	0.00794	54 6	32	0.00063	65 A			



2. Alignment to Reference Genome (BAM/CRAM)



- Align short reads to a known reference (e.g., GRCh38).
- Tools: BWA-MEM, Bowtie2, minimap2.
- Output: SAM/BAM/CRAM file (binary alignment).
- Why? To know where each read comes from in the genome.

Challenges:

- Reads mapping equally well to multiple locations.
- Indels near repetitive regions.

131	141	151	161	171	181	191	
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCCTGGC							← reference
.....							
CATC		CCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGC				CTGGC	
CATCTT		TCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTC				(	
CATCTTCAACACCCGCC		CCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCC					
CATCTTCAACACCCGCCTCT		GCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCCTGG					
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATAT				TCAAGCGAACCACAGTCAAAACGCCCTCCTGGC			
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTA				CGAACCACAGTCAAAACGCCCTCCTGGC			
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAA				ACCACAGTCAAAACGCCCTCCTGGC			
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAA				ACAGTCAAAACGCCCTCCTGGC			
CATCTTCAA					GTCAAAACGCCCTCCTGGC		
CATCTTCAACACC						CCCTCCTGGC	
CATCTTCAACACCCGCCTCTC							
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAA							
	TCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACC						
	CTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCA						← read

SAM/BAM file format

Sorting order of alignments: coordinate

Header	@HD	VN:1.5	SO:coordinate	Header line		Example BAM file	
	@SQ	SN:chr1	LN:248956422	Reference sequence name. Reference sequence length			
	@SQ	SN:chr2	LN:242193529	Read group. Read group identifier. Library. Platform. Sample			
	...						
	@RG	ID:Case_003_T	LB:Case_003_T	PL:ILLUMINA	SM:Case_003_T		
Reads	@PG	ID:bwa	PN:bwa	VN:0.7.17-r1188	CL:bwa mem -t 10 -M -R	...	
	Read1	99	chr1	10147	40	10M	= 10395 310 ACCCTAACCC CHJ<GGIF;G RG:Z:Case_003_T
	Read2	83	chr1	10178	22	7M6H3M	= 10201 -65 CCCTATCCCG F=(2B9DB9) RG:Z:Case_003_T
Read name		Position		Sequence		Base quality	
Mapping quality							

Program. Program record identifier. Program version Program name. C

Base quality scores are per-base estimates of error emitted by the sequencing machines.
Short variant calling algorithms rely heavily on base quality scores

SAM/BAM file format

@HD VN:1.5 S0:coordinate @SQ SN:ref LN:45											Header section
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*	Alignment section
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*	
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; * meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.

RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column.

CIGAR: summary of alignment, e.g. insertion, deletion

MAPQ: mapping quality

POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

QNAME: query template name, aka. read ID

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!-?A-Z]{1,254}	Query template NAME
2	FLAG	Int	[0,2 ¹⁶ -1]	bitwise FLAG
3	RNAME	String	* ! !-()+-<->- !-~ *	Reference sequence NAME
4	POS	Int	[0,2 ³¹ -1]	1-based leftmost mapping POSition
5	MAPQ	Int	[0,2 ⁸ -1]	MAPping Quality
6	CIGAR	String	* ! ([0-9]+[MIDNSHPX=])+	CIGAR string
7	RNEXT	String	* ! !-()+-<->- !-~ *	Ref. name of the mate/next read
8	PNEXT	Int	[0,2 ³¹ -1]	Position of the mate/next read
9	TLEN	Int	[-2 ³¹ +1,2 ³¹ -1]	observed Template LENgth
10	SEQ	String	* ! [A-Za-z=]+	segment SEQUENCE
11	QUAL	String	[!-~]+	ASCII of Phred-scaled base QUALity+33

Base quality correction

- Essential for estimating genotype likelihood
- Context can affect base quality, e.g.:
 - homopolymers
 - cycle
- Estimated error rate $\neq$ 'real' error rate
- Base quality score recalibration (BQSR) takes this context into account



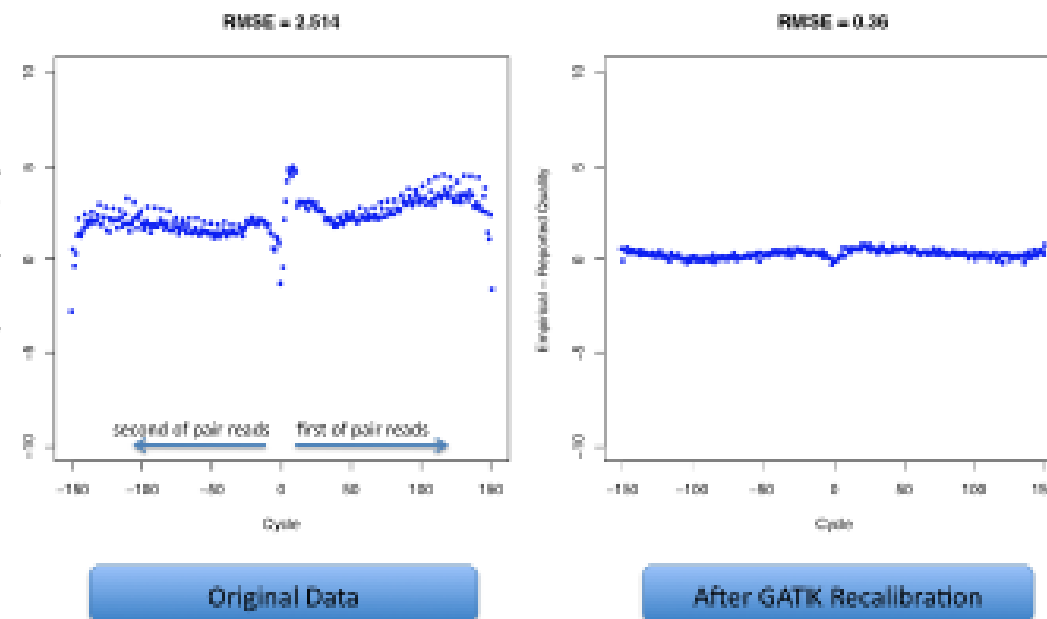
Base Quality Score Recalibration (BQSR)

Base quality scores produced by the machines are subject to various sources of systematic (non-random) bias from library preparation, sequencing, manufacturing defects in the flowcell chips, sequencer variation, and sequencing chemistry.

It results in over- or underestimation of quality scores

BQSR = model errors empirically, analyze the covariates (mainly sequence context and position in read) and adjust the quality scores

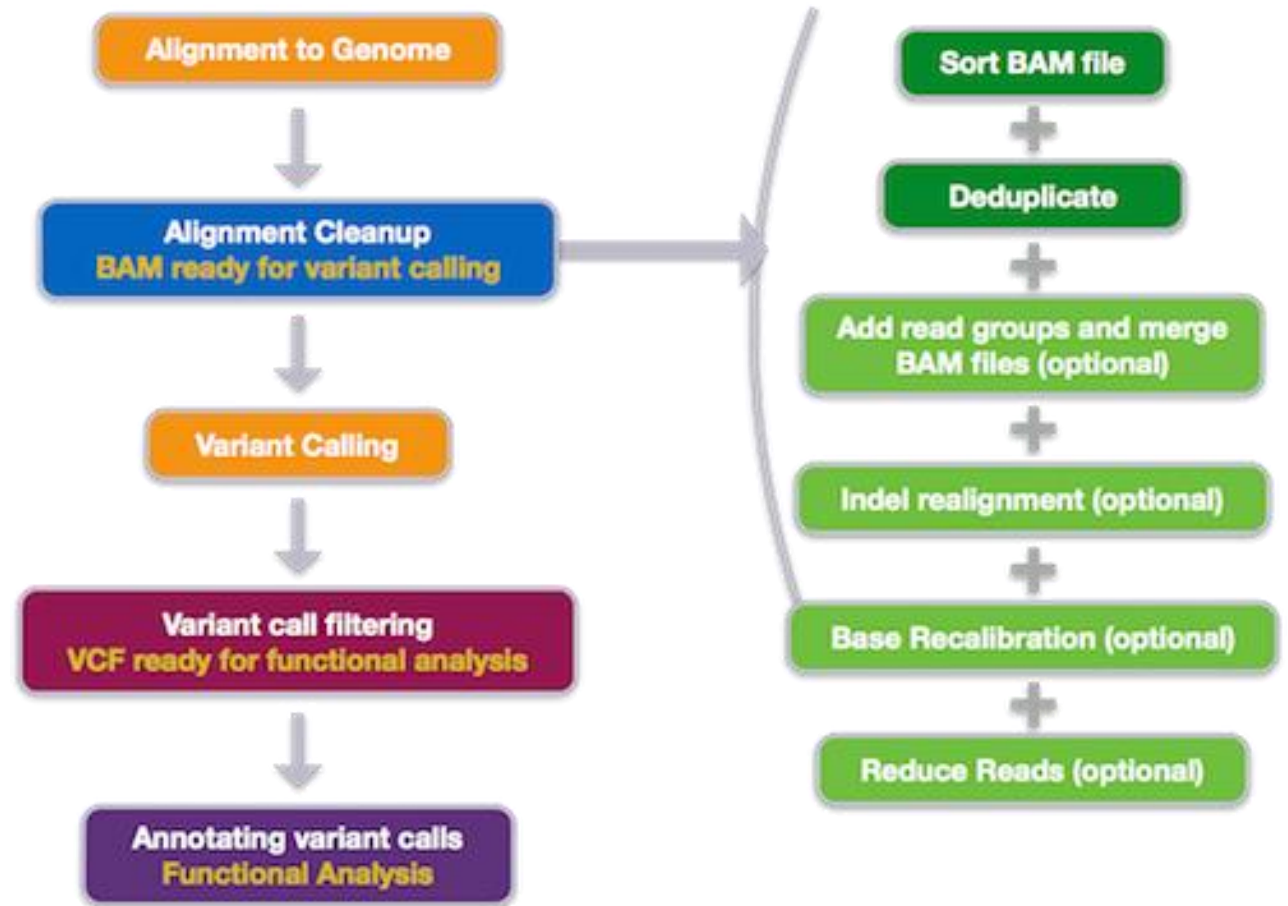
Residual Error by Machine Cycle



3. Preprocessing

Before calling variants, BAM files are "cleaned":

- Sorting: order reads by coordinate.
- Marking duplicates: PCR duplicates can bias allele counts.
- Base Quality Score Recalibration (BQSR): adjust quality scores based on systematic errors.

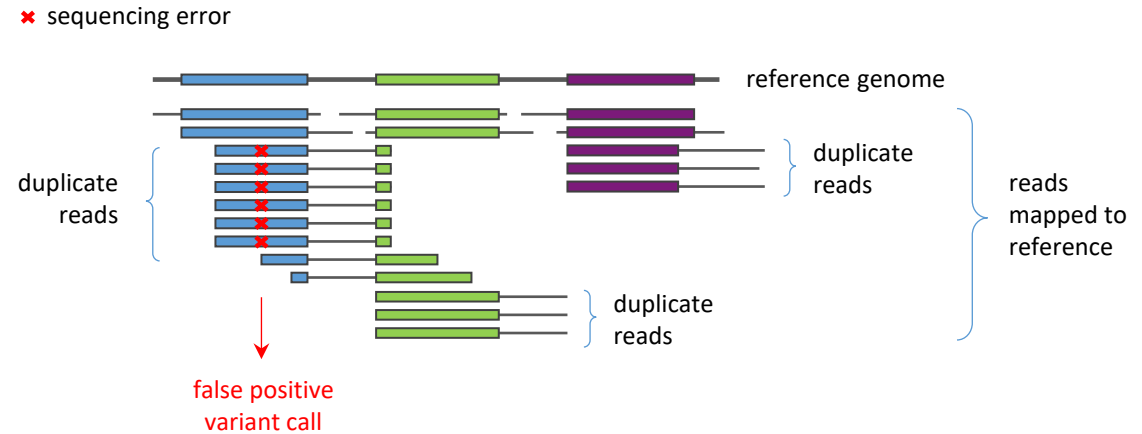


Marking/removing duplicates

Duplicate reads – reads that came from a single DNA fragment.

2 types of duplicates

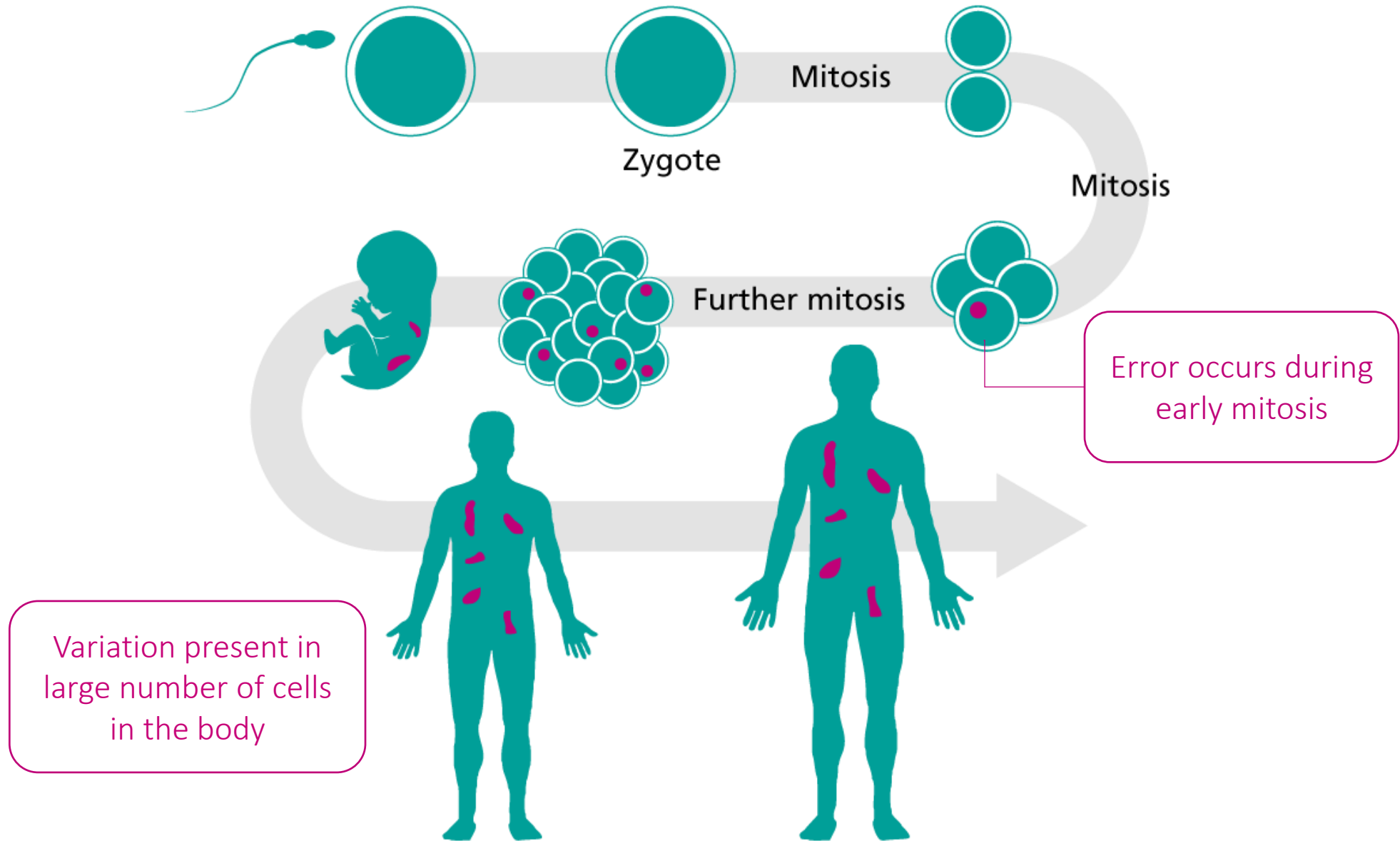
- PCR duplicates (during library preparation)
 - Optical duplicates (a single amplification cluster incorrectly detected as multiple)
-
- Variant calling: each read is an independent observation of the genome
 - Duplicates (can) have the same molecular origin -> not independent
 - In a high-quality library, removing duplicates probably doesn't have a big effect on variant analysis



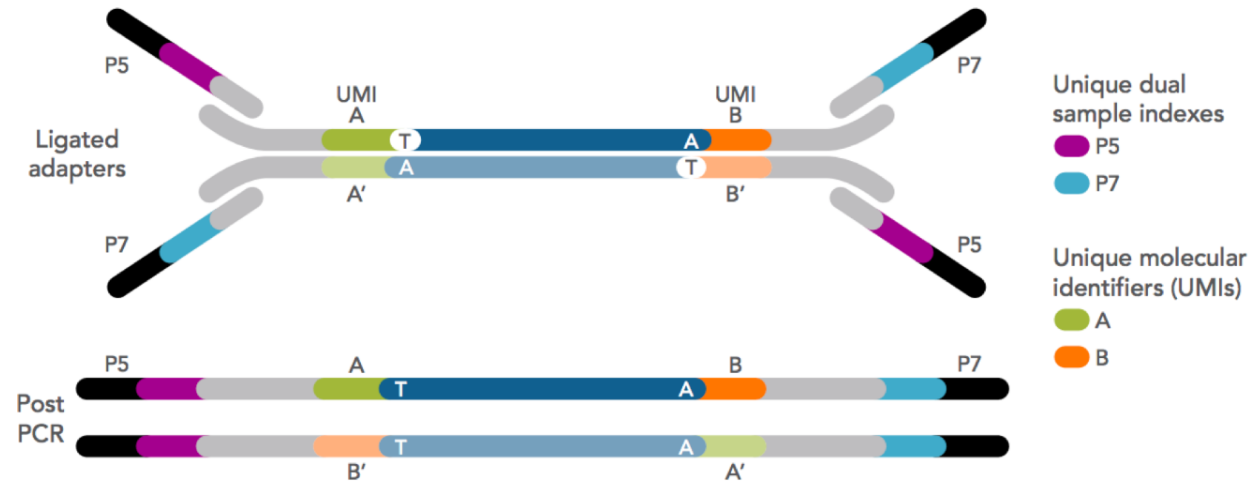
After flagging duplicates, it is more likely that the right call will be made



Somatic mosaic mutations

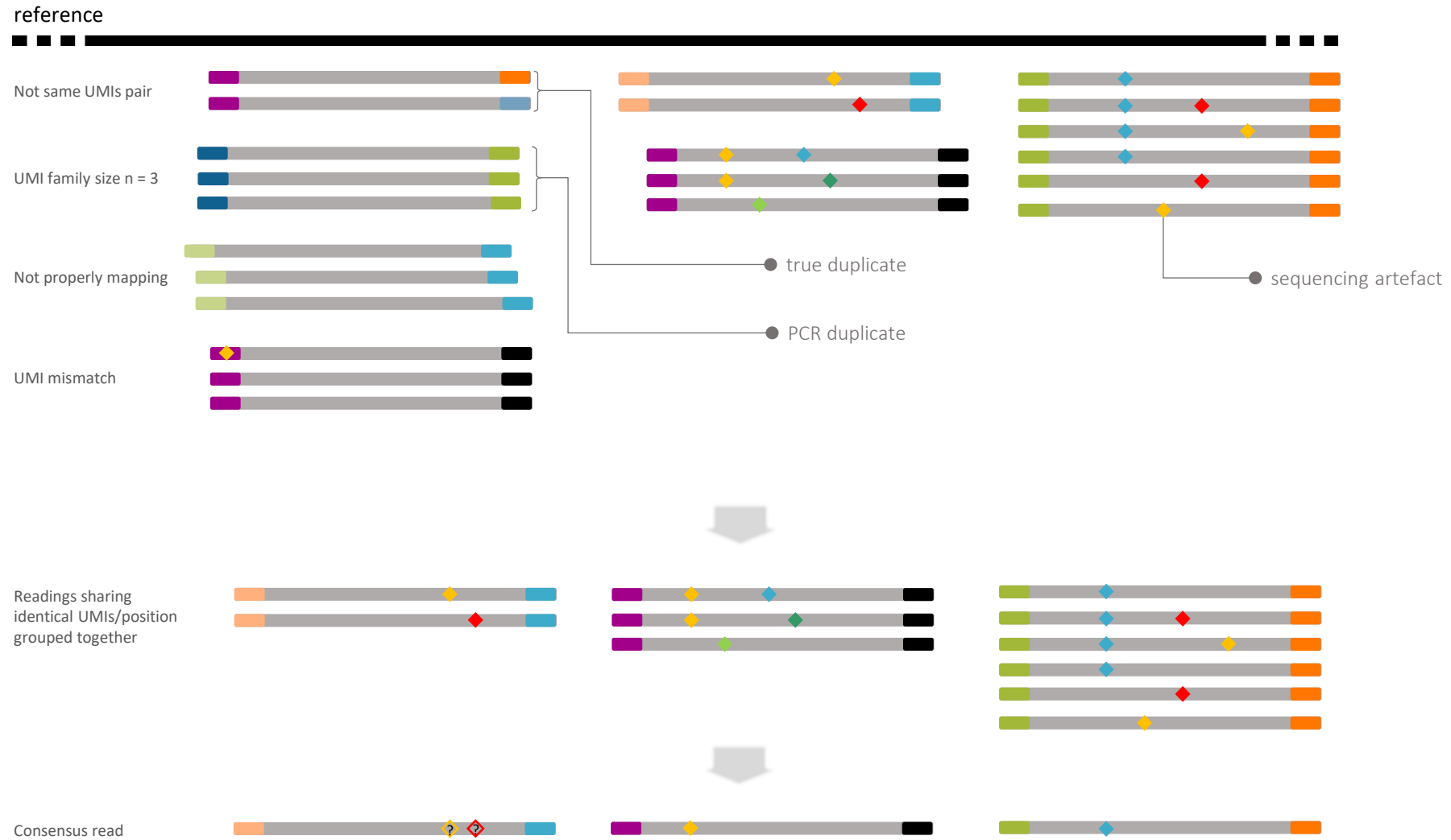


UMIs (Unique Molecular Identifiers)



- Particularly with **low allele frequency** variants and the inherent errors associated with **high-throughput sequencing**
- Improved **differentiation** of **true sequence** variants from **PCR duplicates** and **sequencing artefacts**
- UMI added before PCR reaction
- Detect PCR duplicates and PCR errors

Consensus read / error correction

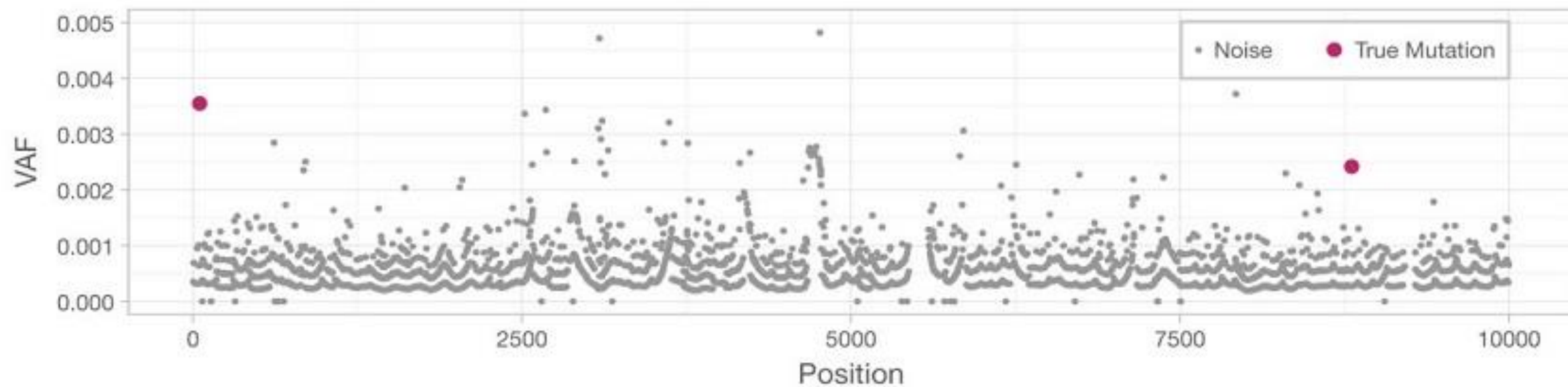


Umi pipeline

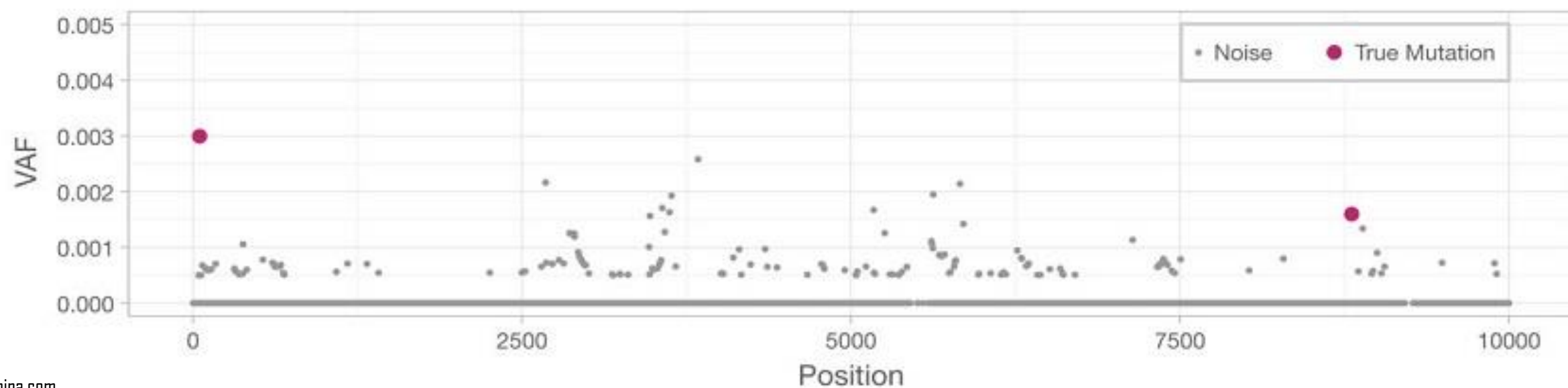
UMI pipeline



Without UMI



With UMI



4. Variant Calling

Now we look for mismatches compared to the reference.

Two main strategies:

- Pileup-based (bcftools, samtools): compare base counts at each position.
- Haplotype-based (GATK HaplotypeCaller, FreeBayes): reconstruct local haplotypes and align reads to them.

Output: VCF (Variant Call Format) — a text file listing all variants.



Questions

1. Is there a variant at location X?
Deviation from REF in the alignments
2. What are the alleles?
The variation in sequence in these deviations
3. What is the genotype (HomRef, Heterozygote or HomAlt)?
Estimating the allele counts in the sample



Variant calling

The goal of somatic variant calling is to identify variants **present in the tumor but not in the matched normal (and are not artefacts)**.

Variants callers are statistical models that typically consider attributes such as base quality, alignment quality and strand to infer the most likely genotype.

Sequencing both tumor and matched normal samples is best for the detection of somatic mutations.

The Statistical Nature of Variant Calling

Variant calling is fundamentally a statistical process, not a simple pattern match. It's not enough to find a mismatch between the sequenced DNA and the reference genome; the challenge lies in evaluating the evidence for that mismatch and deciding whether it represents a genuine variant or is merely a sequencing artifact. This probabilistic approach is critical because all sequencing technologies have an inherent error rate.

The process relies on three primary sources of evidence to distinguish signal from noise:

- **Sequencing Depth (Coverage)**

This refers to the number of times a specific position in the genome has been sequenced. It's a key measure of data quality. A higher coverage provides stronger statistical evidence for a variant. For example, if you see a mismatch at a position that has been sequenced only twice, the confidence is low. If that same mismatch appears in 100 out of 100 reads, the confidence is exceptionally high.

- **Base Quality Scores**

Each base call from a sequencer comes with a Phred score, a logarithmically scaled measure of the confidence in that specific base call. A score of Q30 means there's a 1 in 1,000 chance of the base being wrong. A variant call that is supported by reads with high base quality scores is much more trustworthy than one supported by reads with low-quality scores.

- **Allele Frequency**

This is the proportion of sequencing reads that show the variant allele at a given position. In a standard germline sample, this value is expected to be close to 0% (reference allele), 50% (heterozygous variant), or 100% (homozygous variant). In a tumor sample, however, the allele frequency can be much lower due to the presence of normal cells and sub-clonal populations of cancer cells.

Germline Variant Calling

In germline variant calling, the primary goal is to identify a change that is present in all, or nearly all, of the sequenced cells. The expectation is that the variant will be present in roughly 50% of the reads for a heterozygous variant (one copy of the allele) or 100% of the reads for a homozygous variant (two copies of the allele).

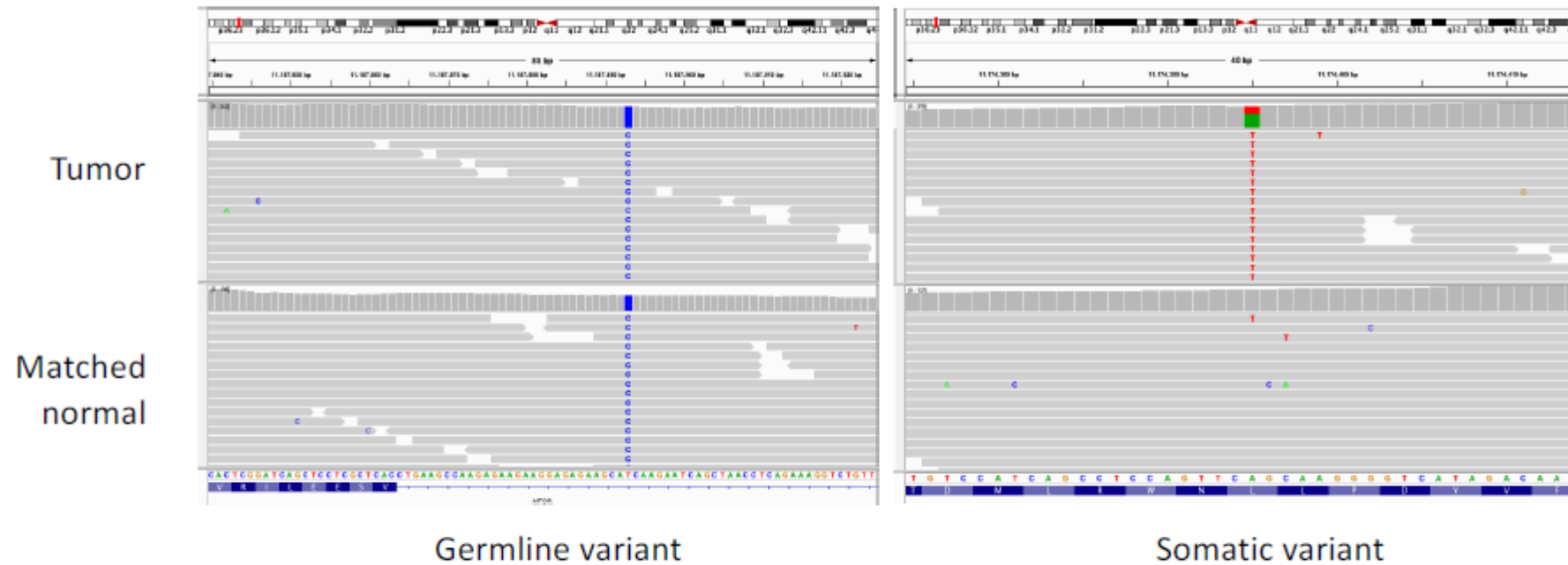
- **Input Data:** Typically uses DNA from a single healthy tissue sample (e.g., blood).
- **Core Assumption:** The variants are constitutional, meaning they are present in all cells.
- **Example Tool:** GATK's HaplotypeCaller is a widely used algorithm specifically designed for this purpose.

Somatic Variant Calling

Somatic variant calling is significantly more complex because it's looking for changes that are present in only a subset of the sequenced cells. For example, in a tumor sample, the cancer cells with the mutation are mixed with normal, non-cancerous cells. This means the variant allele frequency can be very low, well below the 50% or 100% threshold of germline variants.

- **Input Data:** Requires a matched "tumor-normal" pair. The tumor sample is the primary source of the cancer variants, while a normal sample (e.g., blood or adjacent tissue) from the same individual serves as a crucial control to filter out germline variants.
- **Core Assumption:** Variants are sub-clonal and may be present at very low frequencies, so the signal must be distinguished from random sequencing noise.
- **Key Challenge:** Differentiating a low-frequency somatic variant from a low-quality sequencing artifact is a major statistical hurdle.
- **Example Tool:** Mutect2 is a common algorithm designed to handle this complexity by comparing the tumor and normal samples to identify changes unique to the tumor.

Sequencing both tumor and matched normal samples is best for the detection of somatic mutations.



germline variants >> # somatic variants

But often, matched normal samples

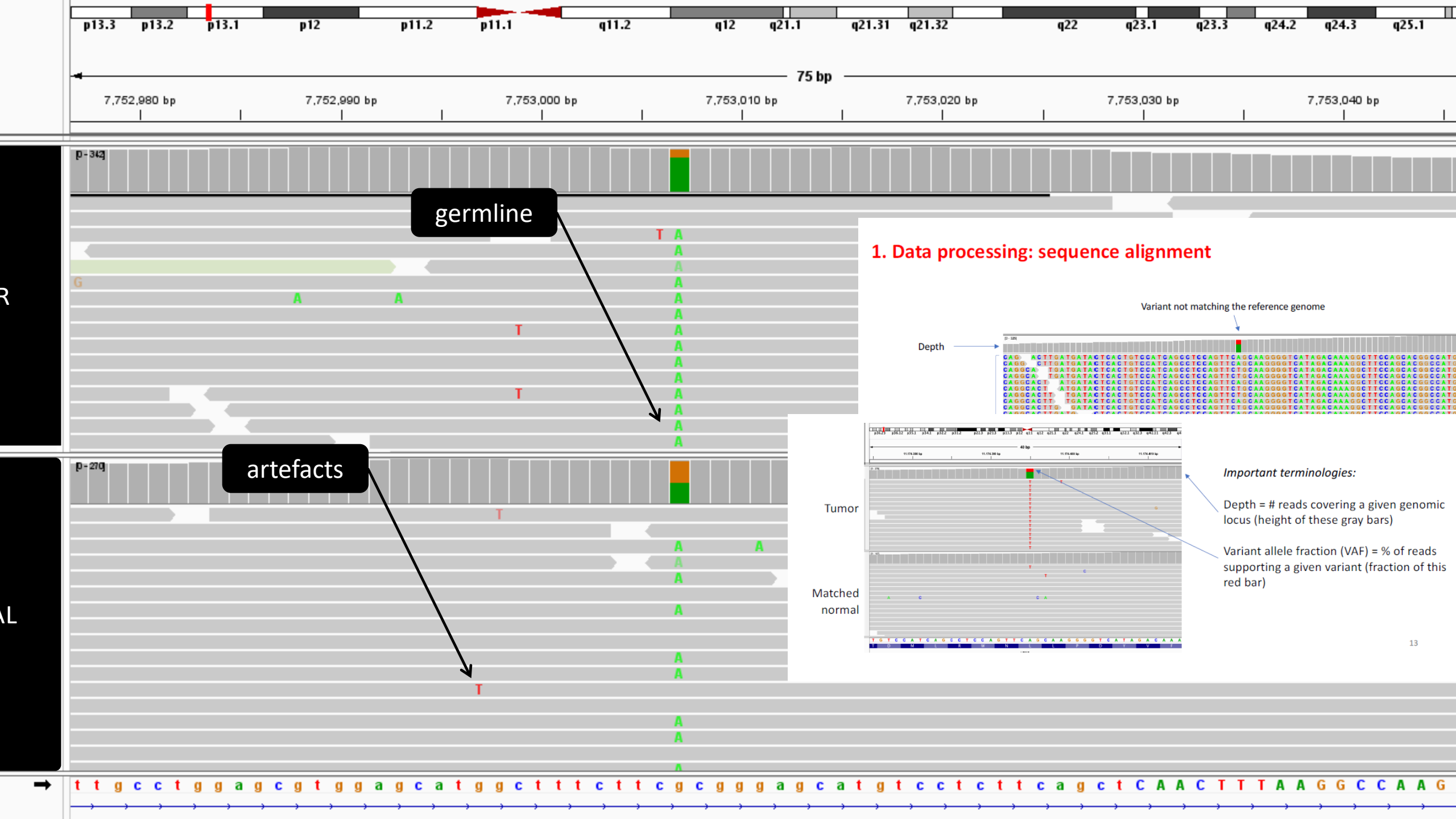
Tumor + matched normal samples

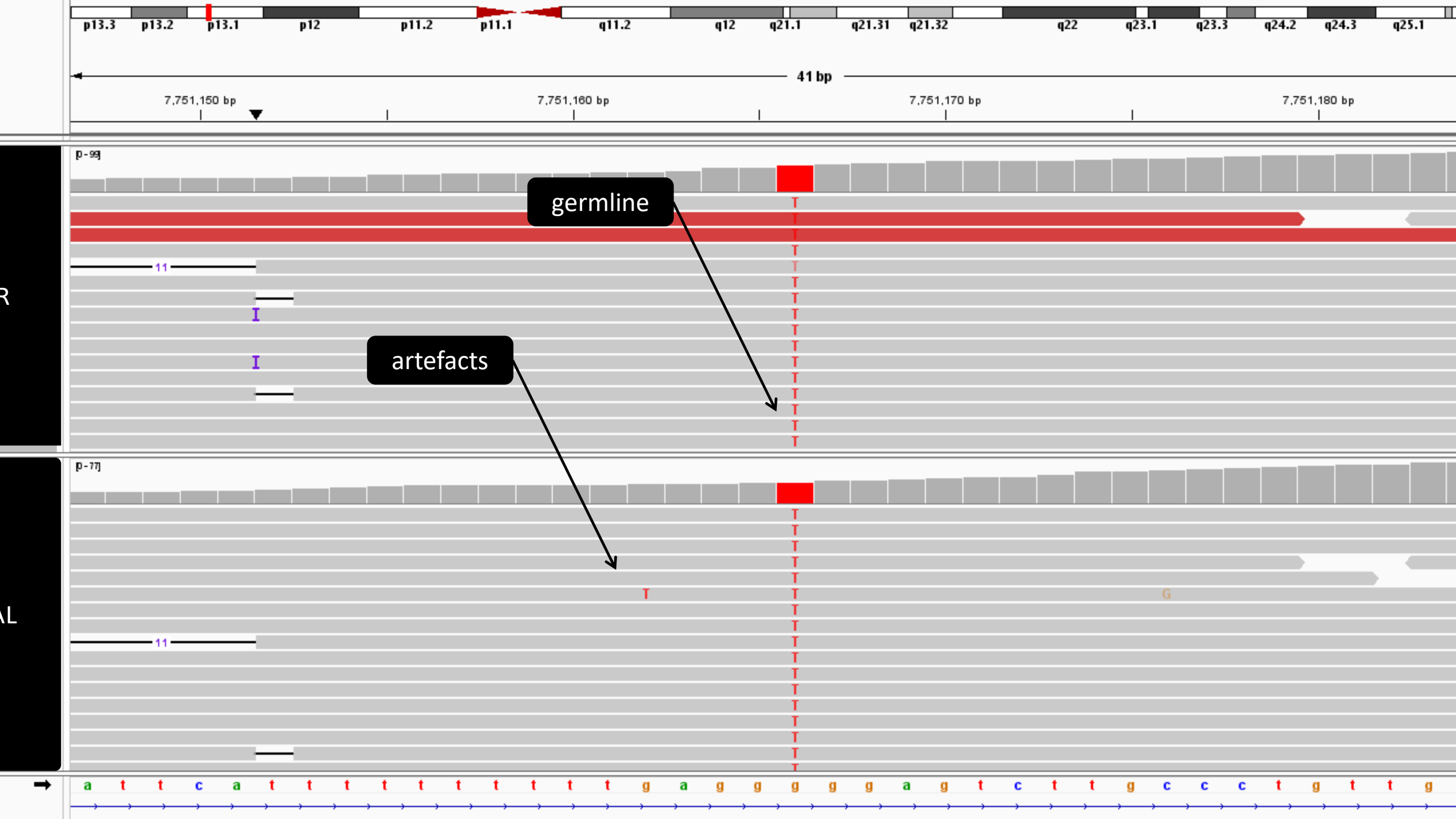
Somatic variant caller that jointly tests T & N

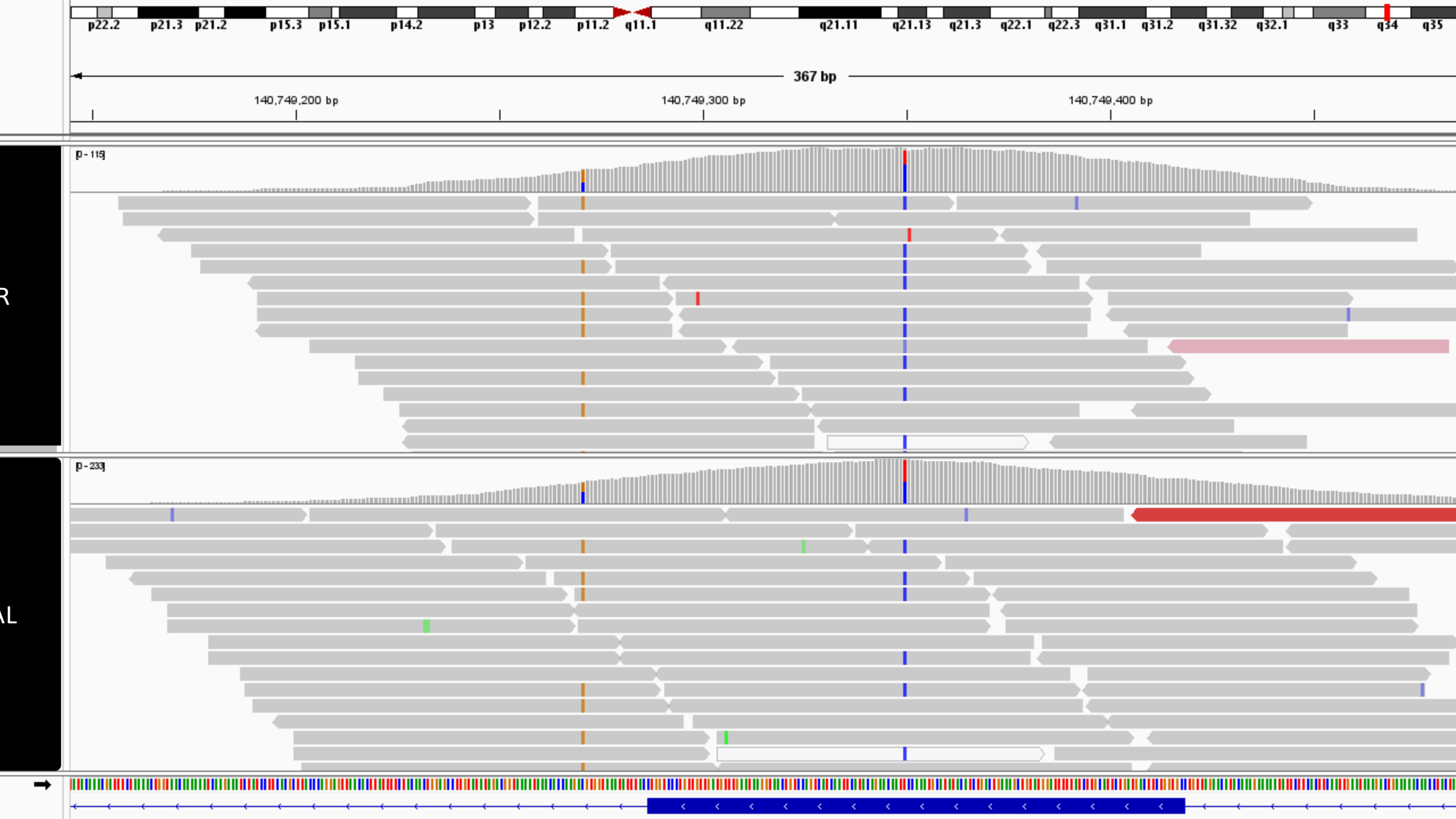
Tumor samples only

Somatic variant caller that jointly tests T & (pool of) unmatched N

Single-sample variant caller that tests T alone (e.g. those used for germline variant calling)







Variable VAF in tumor

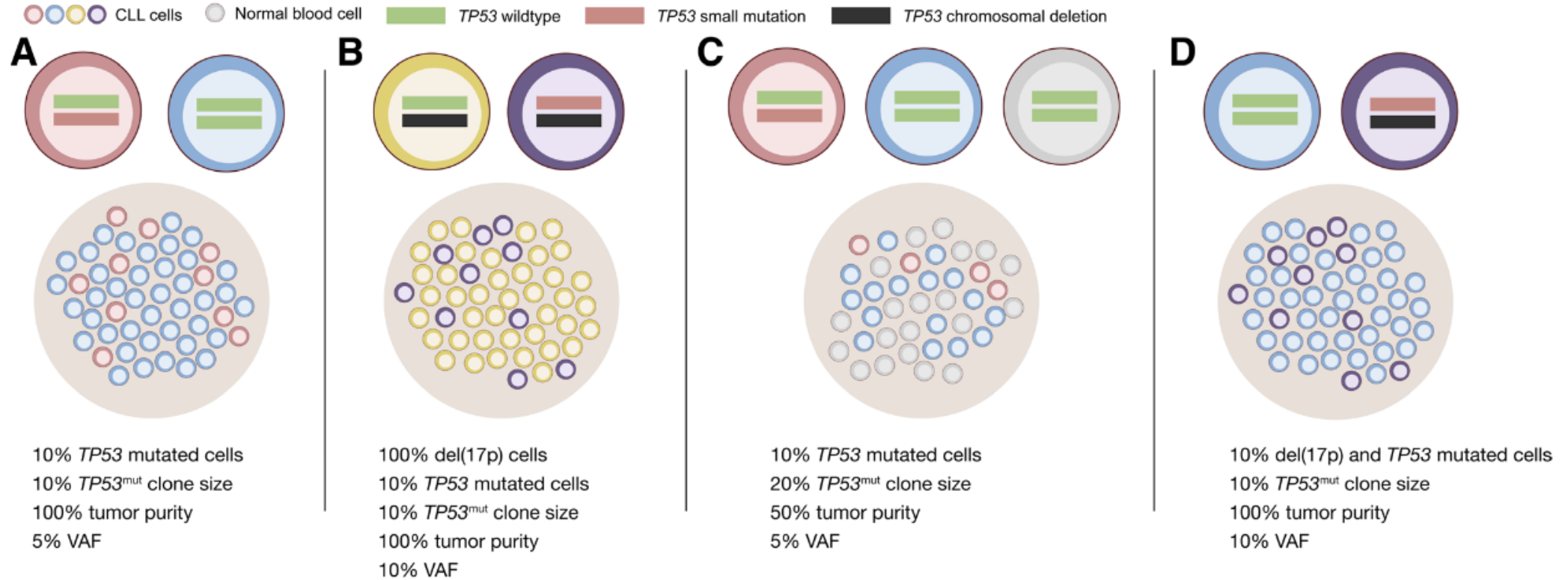
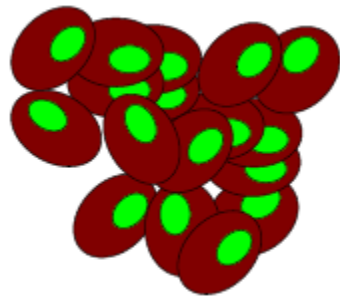
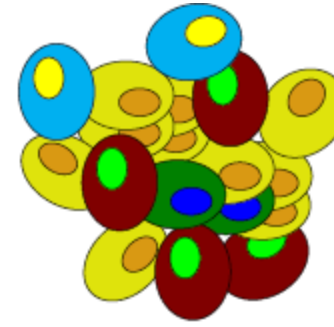


Figure 4. VAF and proportion of cells. (A) In a sample with 100% tumor purity and no del(17p), a *TP53* mutation with VAF 5% (red cells) will correspond to a clone size of 10%. (B) A sample with 100% purity and a del(17p) (yellow cells) will lead to loss of the wild-type allele and increase the *TP53* mutation VAF to 10% with a clone size of 10% (purple). (C) In a less pure sample with 50% tumor cells (blue and red cells), a *TP53* mutation VAF of 5% will correspond to 10% of all cells being mutated but a clone size of 20%. (D) A sample with 100% tumor purity with 10% of the tumor cells having coexisting *TP53* mutation and del(17p) (purple) will correspond to a clone size and VAF of 10%. CLL cells are depicted in blue (no *TP53*^{ab}), yellow [del(17p)], no small *TP53* mutation) red [small *TP53* mutation, no del(17p)] or purple [small *TP53* mutation and del(17p)]. Normal cells are depicted in grey.



Non-tumor cells

Variants are usually present in ~100% or ~50% of the reads for diploid genomes



Tumor cells

Variants can be present in any number (or %) of reads

Intra-tumor heterogeneity (multiple distinct tumor clones)

- Subclonal (not in all tumor cells) mutations have low VAF

Tumor purity = tumor cells/all cells

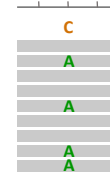
- Tumors contaminated with normal cells may have variants of low VAF
- Tumors with lots of infiltrating cells (e.g. stromal and immune cells) may have variants of low VAF

Genotype

The Model: Binomial Distribution

This scenario perfectly fits a binomial model.

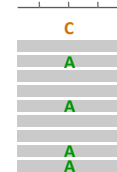
- **Number of trials (n):** The total number of reads sequenced at this position. Here, $n = 9 = 5(\text{ref}) + 4(\text{alt})$
- **Number of successes (k):** The number of times we observe the alternate allele. Here, $k=4$.
- **Probability of success (p):** The theoretical probability of observing the alternate allele, given the model. For a diploid, heterozygous individual, we have one reference allele and one alternate allele, so the theoretical probability is $p=0.5$.



The formula for the binomial probability is:

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k} = \frac{n!}{k!(n - k)!} \cdot p^k \cdot (1 - p)^{n-k}$$

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k} = \frac{n!}{k!(n - k)!} \cdot p^k \cdot (1 - p)^{n-k}$$



It asks: "What is the chance of seeing exactly 4 alternate reads out of 9 total reads, if the true genotype is heterozygous?"

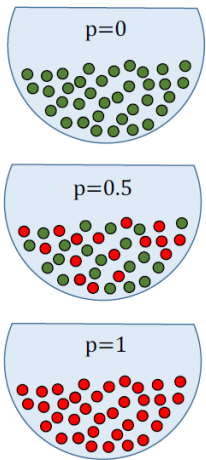
$$P(X = 4) = \binom{9}{4} \cdot 0.5^4 \cdot (1 - 0.5)^{9-4} = 126 \cdot 0.5^4 \cdot 0.5^5 \approx 0.246$$

This is the probability of the observed data given the assumed model.

The probability of observing 4 alternate reads out of 9, assuming a true heterozygous variant, is approximately 24.6%.

Likelihood

In this context, **likelihood is the probability of the model given the observed data**. The numerical value is the same as the probability we just calculated, but the interpretation is different. While probability answers, "What is the chance of this data occurring?" (given a known model), likelihood answers, "How well does this model explain the data?" (when the model is unknown). Variant callers don't just calculate one likelihood; they calculate the likelihood for multiple possible genotypes (models) and then pick the one with the highest value.



- **Likelihood (Homozygous Reference):** What is the chance of observing 4 alternate reads if the true genotype is reference/reference? ($p = 0.0$) -> Likelihood is 0.
- **Likelihood (Homozygous Alternate):** What is the chance of observing 4 alternate reads if the true genotype is alternate/alternate? ($p = 1.0$) -> Likelihood is 0.
- **Likelihood (Heterozygous):** What is the chance of observing 4 alternate reads if the true genotype is heterozygous? ($p = 0.5$) -> Likelihood is approximately 0.246.
- In this case, the likelihood of the heterozygous model being the correct one, given the data, is 0.246. Since this is the highest likelihood among the common genotypes, the variant caller would likely conclude that the genotype is indeed heterozygous.

The formula for likelihood is the same mathematical function as the probability formula, but the interpretation and what you are solving for are different. The general formula for the likelihood of a parameter θ , given some observed data x , is:

$$L(\theta|x) = f(x|\theta)$$

probability mass function (or density function) that you use to calculate the probability of the data. For variant calling, this is the binomial probability formula.

likelihood function. It represents how plausible a specific model parameter θ is, given your observed data x .

Applying the Formula to Variant Calling In your example, the model parameter θ is the true allele frequency (p) of the variant. The observed data x is the number of alternate reads (k) out of the total number of reads (n). So, the likelihood formula for a variant call is the same as the binomial probability formula:

$$L(p|k, n) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

Here's the key difference in interpretation:

- **Probability:** You assume the true genotype is heterozygous ($p=0.5$). You then calculate the probability of observing 4 alternate reads out of 9.
- **Likelihood:** The observed data is fixed (4 alternate reads out of 9). You then use the likelihood formula to compare **different genotype models**. The variant caller calculates the likelihood for:
 - Heterozygous Model ($p=0.5$): $L(0.5|4,9)$
 - Homozygous Reference Model ($p=0$): $L(0|4,9)$
 - Homozygous Alternate Model ($p=1$): $L(1|4,9)$

The variant caller then selects the genotype with the highest likelihood. In the previous example, the likelihood of the heterozygous model (0.246) was the highest, making it the most probable genotype given the evidence.

Estimating the Genotype

This formula calculates the likelihood of a given genotype ($g = 0,1,2$) based on the observed reads, but it adds a crucial layer of complexity by explicitly modeling the sequencing error rate (e).

$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^l [(m - g)\epsilon_j + g(1 - \epsilon_j)] \prod_{j=l+1}^k [(m - g)(1 - \epsilon_j) + g\epsilon_j]$$

where

- $\mathcal{L}(g)$ • likelihood of the assumed genotype g . This is the value we're trying to find
- m • ploidy For a human genome, this is almost always $m=2$ (diploid)
- k • total number of reads at this position
- number of reads with the reference allele
- number of reads with the alternate allele
- l • sequencing error rate. This is the probability that a single sequenced base is incorrect.
- $k - l$
- ϵ

The Logic of the Formula

The formula is a product of probabilities, meaning it calculates the likelihood by multiplying the individual probabilities of each observed read. It's essentially saying: "What is the probability of observing this exact set of k reads, given my assumption about the true genotype (g)?" The formula can be broken down into three main parts:

$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^l [(m - g)\epsilon_j + g(1 - \epsilon_j)] \prod_{j=l+1}^k [(m - g)(1 - \epsilon_j) + g\epsilon_j]$$

This term is a normalization constant that is often dropped in practice because it is constant for a given position and doesn't change the relative likelihoods of different genotypes. It ensures the sum of all probabilities equals one, but it's not the core of the biological calculation.

product for reference reads

product for alternative reads

$$(m - g)\epsilon_j$$

This is the probability of an **error** creating a reference read.
error rate

is the number of non-reference alleles in our assumed genotype. For a heterozygous genotype ($g=1$), $m-g=1$, so this term represents the probability of an error on the non-reference allele.

$$g(1 - \epsilon_j)$$

This is the probability of a **correct read** creating a reference read.

is the number of reference alleles in our assumed genotype. For a heterozygous genotype ($g=1$), this represents the probability of correctly reading the single reference allele.

is the probability of a correct read

$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^l [(m - g)\epsilon_j + g(1 - \epsilon_j)] \prod_{j=l+1}^k [(m - g)(1 - \epsilon_j) + g\epsilon_j]$$

k-l alternate reads
product for alternative reads

$(m - g)(1 - \epsilon_j)$ The probability of a correct read showing an alternate allele (where the genotype is not fully reference).

$g\epsilon_j$ The probability of an error showing an alternate allele (where the genotype is fully reference).

Simplified Analogy

Think of it like this: A variant caller uses this formula to perform a series of thought experiments.

Assume the true genotype is heterozygous ($g=1$) and the error rate is 1%.

- For each reference read: The formula asks, "What's the probability of this being a reference read?" It could be a correct read from the reference allele (99% chance), or a sequencing error on the alternate allele (1% chance). The formula combines these.
- For each alternate read: The formula asks, "What's the probability of this being an alternate read?" It could be a correct read from the alternate allele (99% chance), or a sequencing error on the reference allele (1% chance).

The formula calculates this for every single read and then multiplies them all together to get the final likelihood score for that one genotype. The caller then repeats this for all other possible genotypes (e.g., homozygous reference, homozygous alternate) and chooses the genotype with the highest likelihood.

The beauty of this formula is that it directly models sequencing errors, making it far more accurate than a simple binomial model that assumes no errors. The better the error model, the more reliable the variant calls.

--- Genotype Likelihood Calculations ---

Total Reads (k): 9, Ref Reads (l): 8, Alt Reads: 1

Error Rate (e): 0.01

Likelihood (g=0, Homozygous Ref): 9.2274e-03

PL Score (g=0): 20.35

Likelihood (g=1, Heterozygous): 1.9531e-03

PL Score (g=1): 27.09

Likelihood (g=2, Homozygous Alt): 9.9000e-17

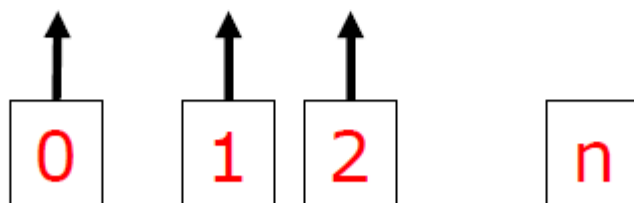
PL Score (g=2): 160.04

Note: The normalization term $1/m^k$ is omitted as it does not affect relative likelihoods.

VCF File Structure

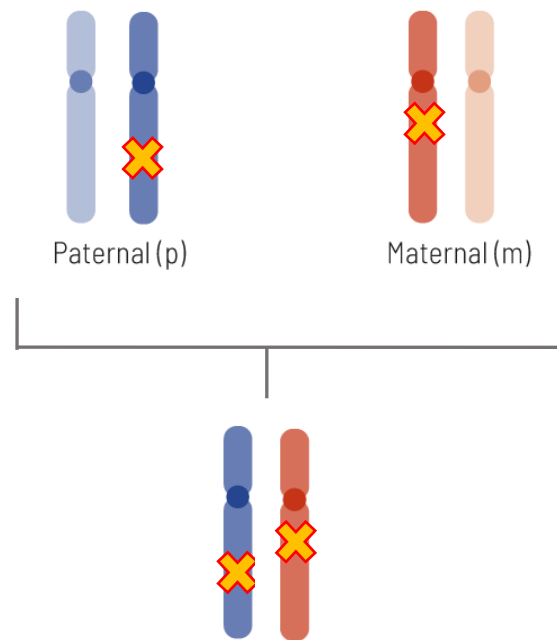
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##fileformat=VCFv4.1
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
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##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

#CHROM	POS	ID	REF	ALT	FORMAT	NA000001	NA000002
20	14370	.	G	A	GT:GQ	0 0:48	1 0:48
20	17330	.	T	A	GT:GQ	0 0:49	0 1:99
20	1110696	.	A	G,T	GT:GQ	1 2:21	2 1:27
20	1230237	.	T	.	GT:GQ	0 0:54	0 0:48
20	1234567	.	GTC	G,GTCT	GT:GQ	0/1:35	0/2:17

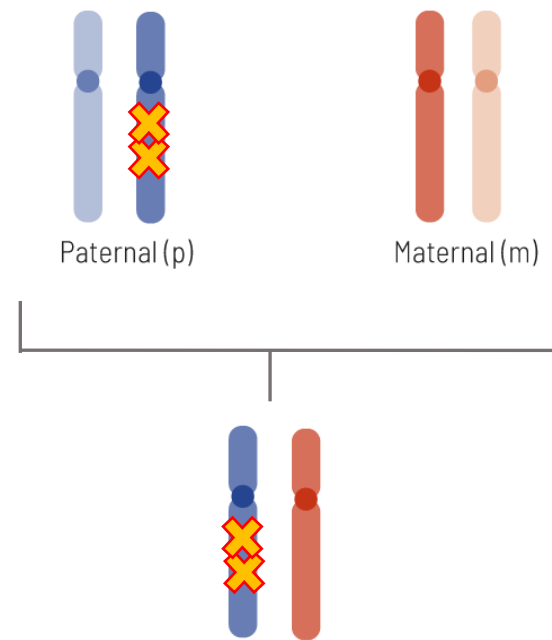


Phasing

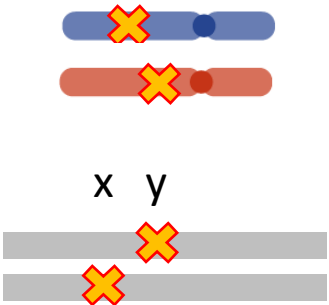
- "Phasing" in genetics is the process of determining which variants on individual parental chromosomes are linked together, essentially creating "haplotypes" or blocks of alleles inherited from one parent. This differentiates a phased genotype, which specifies variants on each chromosome, from an unphased genotype, which only shows the combination of variants at a locus. Phasing is vital for understanding how variants affect an individual's health, particularly for identifying compound heterozygotes and gaining insights into inheritance patterns.
- Why Phasing is Important
 - Clarifying compound heterozygotes:
 - In conditions like certain hereditary cancers, it's crucial to know if two different pathogenic variants are on the same or different parental chromosomes. Phasing clarifies if the variants are on the same strand (cis) or different strands (trans), which is critical for diagnosis.
 - Understanding functional effects:
 - Knowing which variants are inherited together provides information about their combined effects and how they interact.
 - Informing clinical decisions:
 - Phased data can help determine the parental copy carrying a specific deletion, which has implications for an individual's diagnosis and treatment.



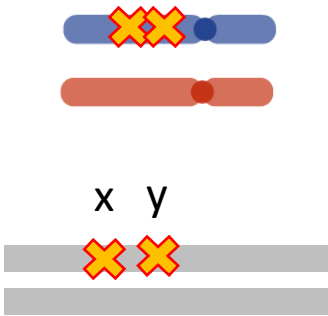
both copies affected
disease



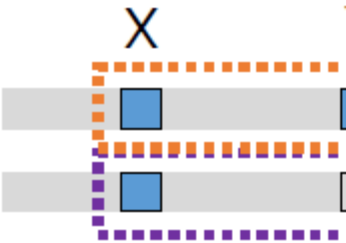
only one copy affected
No disease



CHROM	POS	ID	S1
chr20	101	snp_x	0 1
chr20	204	snp_y	1 0



CHROM	POS	ID	S1
chr20	101	snp_x	1 0
chr20	204	snp_y	1 0

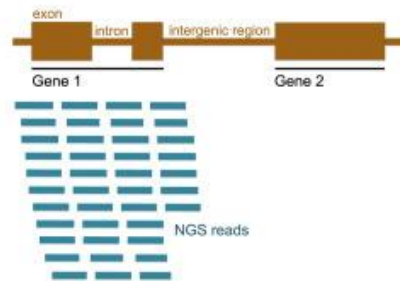


sample1.vcf	
CHROM	POS
20	1101
20	1203

sa

What to sequence?

Targeted Sequencing (TS)

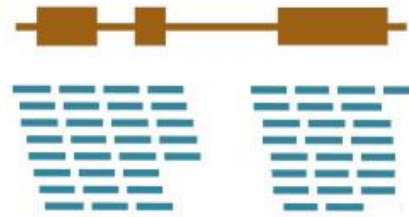


Selection (capture):
Panel of genes of interest
(typically 20-200 predefined genes)

- greatest sequencing depth*
- most cost effective

Fixed selection of genes,
analysis cannot be expanded

Whole Exome Sequencing (WES)

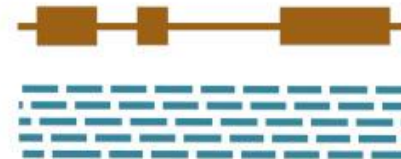


Selection (capture):
All exons of all known genes
(1.5-2% of all human DNA)

- variable read depth at boundaries
- greater sequencing depth*
- more cost effective

Subsequent *in silico* selection of genes/gene list(s) of interest
(but entire dataset generated: analysis can be expanded to other genes)

Whole Genome Sequencing (WGS)



No selection:
Entire human DNA analyzed
Including introns, RNA genes, etc.

- moderate read depth*
- similar read depth throughout
- may also detect copy number variants and repeat expansions
- higher price

Variant calling : basic steps

1. Read mapping / preprocessing

- **Goal:** Align raw sequencing reads to a reference genome.
- **Steps:**
 - Read mapping (e.g., BWA, Bowtie2).
 - Remove duplicates, perform base quality score recalibration.
- **Stats:**
 - Mapping quality (MAPQ) = $-10 \cdot \log_{10} P(\text{read misaligned})$.
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2. Candidate variant detection

- **Goal:** Detect genomic loci where the observed bases differ from the reference more than expected under sequencing error alone.

Statistical approaches used

(a) Base counting and pileup

- At each genomic position, the caller builds a pileup of overlapping reads.
- Each base has:
 - Base quality (Q) → probability base is wrong:

$$P(\text{error}) = 10^{-Q/10}$$

- Mapping quality (MAPQ) → probability read is misaligned.

(b) Binomial test for allele imbalance

- Suppose at a site you have n reads, of which k support an alternate allele.
- If sequencing error rate is ϵ , the null hypothesis is:

$$H_0 : k \sim \text{Binomial}(n, \epsilon)$$

- Compute probability of seeing $\geq k$ alternate reads just by error:

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} \epsilon^i (1 - \epsilon)^{n-i}$$

- If this probability is very small $\rightarrow$ site becomes a candidate variant.

(c) Likelihood-based evaluation in Variant Calling



Goal

Instead of only counting alleles, compute the probability of the observed read data under each possible genotype hypothesis (AA, AB, BB).

This allows:

- Using base quality scores (errors are not uniform).
- Using mapping quality scores.
- Extending naturally to indels and complex variants.

Step 1: Genotype hypotheses

For a diploid organism at a candidate SNP position:

- $g \in \{AA, AB, BB\}$
- Prior allele frequency might come from population data, but at this stage we just compute likelihoods.

Step 2: Read likelihood model

For each read r_i : Let observed base = b_i . Let base quality = Q_i . Error probability = $\epsilon_i = 10^{-Q_i/10}$.

If genotype = AA (homozygous ref): Probability of read i given genotype: $P(r_i|g=AA) = \begin{cases} 1 - \epsilon_i & \text{if } b_i = A \\ \epsilon_i / 3 & \text{if } b_i \neq A \end{cases}$

If genotype = AB (heterozygous): Expect 50% A, 50% B alleles:

$P(r_i|g=AB) = 0.5 \cdot P(r_i|A) + 0.5 \cdot P(r_i|B)$

If genotype = BB: Symmetric to AA.

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If genotype = AB (heterozygous):

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If genotype = BB:

- Symmetric to AA.

1 2
3 4 Step 3: Genotype likelihood Assuming independence of reads:
 $L(g) = \prod_{i=1}^n P(r_i | g)$
 Since products get very small, we use log-likelihoods:
 $\log L(g) = \sum_{i=1}^n \log P(r_i | g)$

1 2 3 4 Step 3: Genotype likelihood

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 Step 4: Genotype assignment Compute likelihoods for all genotypes. Often convert to Phred-scaled likelihoods (PL): $PL(g) = -10 \cdot \log_{10} \left(\frac{L(g)}{\max_g L(g)} \right)$ Caller reports the genotype with lowest PL (highest likelihood).

 Step 4: Genotype assignment

- Compute likelihoods for all genotypes.
- Often convert to **Phred-scaled likelihoods (PL)**:

$$PL(g) = -10 \cdot \log_{10} \left(\frac{L(g)}{\max_g L(g)} \right)$$

- Caller reports the genotype with **lowest PL (highest likelihood)**.

 **Numeric Example** Imagine: Reference = A Candidate variant = G
10 reads covering site: 7 show A (Q=30) 3 show G (Q=30) Error probability:

$\epsilon = 10^{-30/10} = 0.001$
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Genotype AB (A/G): Each read is 50% chance of A or G: For an A read:

$0.5 \cdot (1-\epsilon) + 0.5 \cdot (\epsilon/3) \approx 0.5 \cdot 0.999 + 0.5 \cdot 0.00033 \approx 0.4997$
 $0.5 \cdot (\epsilon/3) \approx 0.5 \cdot 0.00033 \approx 0.000165$. For a G read: same logic, also ≈ 0.4997 .

Genotype GG: 7 A reads: each ≈ 0.00033 3 G reads: each ≈ 0.999

$L(AA) \approx 0.993$
 $L(AB) \approx 0.4997^{10} \approx 9.8 \times 10^{-4}$
 $L(GG) \approx 3.6 \times 10^{-22}$. Result $L(AB) \gg L(AA), L(GG)$. Variant caller assigns genotype = AB (heterozygous). PL values would show: PL(AB) = 0 (best) PL(AA) ≈ 70 PL(GG) ≈ 220

Numeric Example

Imagine:

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Error probability: $\epsilon = 10^{-30/10} = 0.001$.

Likelihoods:

Genotype AA (all A):

- 7 A reads: $(1 - \epsilon)^7 \approx 0.999^7 \approx 0.993$
- 3 G reads: $(\epsilon/3)^3 \approx (0.00033)^3 \approx 3.6 \times 10^{-11}$
- $L(AA) \approx 3.6 \times 10^{-11}$

Genotype AB (A/G):

Each read is 50% chance of A or G:

- For an A read: $0.5 \cdot (1 - \epsilon) + 0.5 \cdot (\epsilon/3) \approx 0.5 \cdot 0.999 + 0.5 \cdot 0.00033 \approx 0.4997$
- For a G read: same logic, also ≈ 0.4997 .
- $L(AB) \approx 0.4997^{10} \approx 9.8 \times 10^{-4}$.

Genotype GG:

- 7 A reads: each ≈ 0.00033
- 3 G reads: each ≈ 0.999
- $L(GG) \approx (0.00033)^7 \cdot (0.999)^3 \approx 3.6 \times 10^{-22}$.

(d) Strand bias and read orientation Sequencing errors are often biased to one strand. Fisher's Exact Test or a binomial test is used: Compare counts of alternate alleles on forward vs reverse strand. Null: alleles should be equally distributed between strands.

(e) Indel realignment / local assembly For indels, mismatches may just be due to misalignment. Candidate indels are often detected by: Clusters of soft-clipped reads. Realignment and re-scoring likelihoods using pair-HMMs (used in GATK HaplotypeCaller, Mutect2).

(f) Practical thresholds (heuristics) Depth filters (minimum coverage). Alternate allele count thresholds (e.g., ≥ 2 alt reads). Allele fraction filters (e.g., $\geq 1-2\%$ for somatic; $\geq 20-30\%$ for germline heterozygotes). Often combined with statistical tests above.

Summary of stats at candidate detection Binomial / multinomial models $\rightarrow$ test if alt allele count can be explained by error. Likelihood models $\rightarrow$ genotype likelihoods given read qualities. Fisher's exact test $\rightarrow$ strand bias. Pair-HMMs / assembly $\rightarrow$ handle indels and complex variants.

3. Genotype likelihoods

- **Goal:** Estimate likelihood of each possible genotype (e.g., AA, AB, BB for diploids).
- **Formula** (simplified):

$$L(g) = \prod_i P(\text{base}_i | g, q_i)$$

where q_i = base quality.

- **Stats:**
 - **Bayesian inference:**

$$P(g|\text{data}) \propto P(\text{data}|g) \cdot P(g)$$

- **Priors on allele frequency** (e.g., Hardy–Weinberg, population priors).

4. Variant calling decision

- **Goal:** Decide whether to emit a variant at a site.
- **Stats:**
 - Likelihood ratio test (LRT) comparing best non-reference genotype vs reference.
 - Quality score:

$$Q = -10 \cdot \log_{10} P(\text{variant call is wrong})$$

- Somatic callers (tumor/normal pairs) often use binomial tests, Fisher's exact test, or beta-binomial models to detect allele imbalance between tumor vs normal.

5. Filtering & post-processing

- **Goal:** Reduce false positives.
- **Stats:**
- **Hard filters:** thresholds on depth, strand bias, mapping quality, allele fraction.
- **Strand bias tests:** Fisher's exact test or binomial test.
- **Overdispersion modeling:** beta-binomial (to model extra variance in allele counts).
- **Machine learning** (GATK VQSR): Gaussian mixture models to separate true vs false variants using training sets.